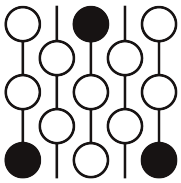
Review of Problem 1 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

The proof is correct and very clearly written. It follows exactly the same strategy as the human solution, but avoids approximation of birhizomic families and therefore achieves a simpler proof. I ranked it third among the solutions I read. The similarity to the human solution made this writeup very easy to understand for a reviewer who was familiar with that solution, but this solution might be more difficult to understand for a reader who is less familiar with the problem. Some basic facts about monoids could have been cited instead of rederived, but since these facts are fairly standard, I do not fault the solution for lack of attributions.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 1

2 Submission with inline comments

Abstract

The answer to the question is affirmative, without a finite-generation hypothesis on the monoid. We prove a stronger statement: the presheaf left Kan extension of the given presheaf is already a birhizomic sheaf and is acyclic on every object of the birhizomic site. The proof uses finite refinements of birhizomic covers, a finite convex-cover calculation, and passage to filtered unions of finitely generated saturated monoids.

3 Statement and conventions

We use the definition of monoid in the question. In intermediate constructions, submonoids of groups are allowed to have units; we explicitly sharpen them before regarding them as objects of the category in the question. All group-valued expressions involving a monoid homomorphism refer to its extension to groupifications.

Put

$$P_x = P[-x]^\sharp \qquad (x \in P).$$

A presheaf on B_P^{op} will be regarded as a covariant functor $B_P \to \mathrm{Ab}$. Likewise, presheaves on A_P^{op} are covariant functors on A_P.

For a sheaf $\mathcal{E}$ on A_P^{op}, write $H^q(Q, \mathcal{E})$ for the right derived functors of evaluation at the object $P \to Q$. We make the usual universe choices for sites and sheaf categories.

Theorem 3.1. *Let P be any monoid, and let $F : B_P \to \mathrm{Ab}$ be any functor. For $f : P \to Q$, define*

$$\mathcal{G}_F(Q) = \operatorname{colim}_{\substack{x \in P \\ f(x)=0}} F(P_x), \tag{1}$$

where the indexing category consists of the corresponding objects P_x and their morphisms over P. Then:

(i) *$\mathcal{G}_F$ is the presheaf left Kan extension of F, and it is already a sheaf for the birhizomic topology;*

(ii) *for every $P \to Q$,*

$$H^q(Q, \mathcal{G}_F) = 0 \qquad (q > 0);$$

(iii) *the canonical unit is a quasi-isomorphism:*

$$F \xrightarrow{\sim} \mathrm{R}\pi_* \pi^{-1} F.$$

The indexing category in (1) is filtered, as will be proved below. In particular, the formula does not require its indexing face to be finitely generated.

4 Elementary constructions with monoids

4.1 Filtered colimits and finite rational extensions

Lemma 4.1. *The groupification of a monoid is torsion-free. Filtered colimits of monoids are computed on the underlying commutative monoids, and hence on the underlying sets.*

Proof. If $ng = 0$ in P^{gp}, with $n > 0$, saturation gives $g, -g \in P$, and sharpness gives $g = 0$.

Let $P = \operatorname{colim}_\lambda P_\lambda$ be a filtered colimit of the underlying commutative monoids. Integrality and sharpness hold in P: any equality witnessing a failure of either property holds at some stage, where that property is valid.

To check saturation, write an element of P^{gp} as $a-b$, with $a, b \in P$, and suppose $n(a-b) = c \in P$. By integrality, this is the equality $na = nb + c$ in P. Representatives for a, b, c, together with this equality, exist at a common stage. Saturation at that stage shows that $a - b$ belongs to its positive monoid, and therefore belongs to P. □

Definition 4.2. For a finite subset $S \subset P^{\mathrm{gp}}$, define

$$C_P(S) = \left\{ g \in P^{\mathrm{gp}} : ng \in P + \sum_{s \in S} \mathbb{N}s \text{ for some } n > 0 \right\},$$

and put

$$P\langle S\rangle = C_P(S)/\big(C_P(S) \cap (-C_P(S))\big).$$

We call $P \to P\langle S\rangle$ a *finite rational extension*. This terminology is used only for the construction just specified.

Lemma 4.3. *The construction in Definition 4.2 has the following properties.*

(i) *For any $h : P \to M$, there is a morphism $P\langle S\rangle \to M$ extending h if and only if $h(s) \in M$ for all $s \in S$. Such a morphism is unique.*

(ii) *The morphism $P \to P\langle S\rangle$ is of finite presentation, and its groupification is surjective.*

(iii) *Finite rational extensions are preserved by arbitrary pushout and by composition. Moreover,*

$$P\langle S\rangle \amalg_P P\langle T\rangle \simeq P\langle S \cup T\rangle.$$

(iv) *$P\langle\{-x\}\rangle \simeq P_x$ for $x \in P$.*

Proof. The submonoid $C = C_P(S)$ is saturated in P^{gp}, and its units form the subgroup $U = C \cap (-C)$. The quotient C/U is saturated: if $n(g + U)$ belongs to C/U, then $ng \in C$, since adding or subtracting an element of U does not change membership in C. Thus $g \in C$. The quotient is integral and sharp by construction.

If $h(s) \in M$ for all $s \in S$, saturation of M implies that $h(C) \subset M$, and sharpness implies that $h(U) = 0$. This gives the asserted extension. Its uniqueness follows because $P^{\mathrm{gp}} \to (P\langle S\rangle)^{\mathrm{gp}}$ is surjective.

Write $s = a_s - b_s$, with $a_s, b_s \in P$. The condition $h(s) \in M$ is equivalent to the existence of an element $z_s \in M$ such that

$$h(b_s) + z_s = h(a_s).$$

There are finitely many such conditions, and each z_s, when it exists, is unique by integrality. Lemma 4.1 therefore shows that $\mathrm{Hom}_P(P\langle S\rangle, -)$ commutes with filtered colimits. This proves finite presentation.

The pushout assertion follows immediately from (i). More generally, under $h : P \to Q$, the pushout is $Q\langle h(S)\rangle$. For composition, lift the finitely many group elements used in a rational extension of $P\langle S\rangle$ to P^{gp}; adjoining these lifts together with S gives the composite.

Finally, $h(-x) \in M$, with $h(x) \in M$, is equivalent to $h(x) = 0$, since M is sharp. This is precisely the universal property of P_x. □

Remark 4.4. Every finite rational extension is an epimorphism in the category of monoids, by the uniqueness assertion in Lemma 4.3(i). Consequently, its opposite is a monomorphism of the site. This observation will allow us to use finite alternating Čech complexes.

4.2 Faces and the category B_P

A *face* of P is a submonoid $E \subset P$ such that $a + b \in E$ implies $a, b \in E$. The inverse image of 0 under a homomorphism of sharp monoids is a face.

Lemma 4.5. *For $x \in P$, set*

$$E_x = \{a \in P : a \leq nx \text{ for some } n \in \mathbb{N}\}.$$

Then $E_x = \ker(P \to P_x)$, and

$$\mathrm{Hom}_P(P_x, P_y) \neq \varnothing \quad \Longleftrightarrow \quad E_x \subset E_y.$$

When this set of morphisms is nonempty, it is a singleton.

If P is finitely generated, every face is of the form E_x, there are only finitely many faces, and B_P, up to equivalence, is their inclusion poset.

Proof. The localization $P + \mathbb{N}(-x) \subset P^{\mathrm{gp}}$ is already saturated. Indeed, if $ng = a - kx$, with $a \in P$ and $k \geq 0$, then

$$n(g + kx) = a + (n-1)kx \in P,$$

so $g + kx \in P$. An element $a \in P$ becomes a unit in this localization precisely when $a \leq nx$ for some n. This proves the kernel formula. The assertion about morphisms follows from the universal property of P_x.

If $p_1, \ldots, p_r$ generate P, a face E is generated by those p_j which belong to E: every summand in an expression for an element of a face belongs to that face. Put

$$x_E = \sum_{p_j \in E} p_j.$$

Every element of E is bounded above by a multiple of x_E, and the face property gives $E_{x_E} \subset E$. Hence $E = E_{x_E}$. The same description proves finiteness of the set of faces. □

4.3 Finite-dimensional approximation

We call a monoid *fs* if it is finitely generated, in addition to the standing assumptions.

Lemma 4.6. *The following assertions hold.*

(i) *Every monoid is a filtered union of fs submonoids.*

(ii) *If P is fs and $S \subset P^{\mathrm{gp}}$ is finite, then $P\langle S\rangle$ is fs.*

(iii) *If P is fs, there is a homomorphism $\ell : P \to \mathbb{R}_{\geq 0}$ which is strictly positive on $P \setminus \{0\}$.*

Proof. We first recall the relevant elementary lattice calculation. Let L be a finitely generated free abelian group, and let $v_1, \ldots, v_r \in L$. If

$$C = \sum_j \mathbb{R}_{\geq 0} v_j,$$

then the saturation in L of $\sum_j \mathbb{N} v_j$ is $C \cap L$. Indeed, a nonnegative real solution to a linear system with rational coefficients and rational right-hand side can be replaced by a nonnegative rational solution. One way to see this is to choose a solution with minimal support: dependence among its supporting columns permits movement until a coordinate becomes zero; with independent supporting columns, the solution is rational by linear algebra. Clearing denominators proves the assertion.

Furthermore, $C \cap L$ is finitely generated. For $g = \sum_j t_j v_j \in C \cap L$, subtract $\sum_j \lfloor t_j \rfloor v_j$. The remainder is a lattice point in the bounded set

$$\left\{ \sum_j u_j v_j : 0 \le u_j \le 1 \right\}.$$

There are only finitely many such remainders, and they, together with the v_j, generate $C \cap L$.

For a finite subset of an arbitrary monoid P, take the saturation of the submonoid it generates in the subgroup it generates in P^{gp}. That subgroup is a finitely generated torsion-free abelian group, hence is free. The saturation lies in P, is finitely generated by the preceding calculation, and is sharp. These submonoids are directed under inclusion and exhaust P. This proves (i). Applying the same calculation to generators of P together with S, and then taking the quotient by units, proves (ii).

For (iii), assume $P \neq \{0\}$, and choose nonzero generators $p_1, \ldots, p_r$. The origin does not belong to their convex hull. Otherwise the rational-solution argument above would give a nontrivial relation

$$\sum_j n_j p_j = 0, \qquad n_j \in \mathbb{N},$$

contrary to sharpness. Choose an inner product on $P^{\mathrm{gp}} \otimes \mathbb{R}$, and let z be the point in this compact convex hull nearest the origin. Then $z \neq 0$, and minimality gives

$$\langle z, p_j \rangle \ge \|z\|^2 > 0$$

for every j. The functional $u \mapsto \langle z, u \rangle$ has the required property. For $P = \{0\}$, the assertion is immediate. □

5 Infinitesimal envelopes and finite covering refinements

5.1 An explicit infinitesimal envelope

Lemma 5.1. *Let $v : P \to V$ be a homomorphism to a valuative monoid. There is a factorization*

$$P \longrightarrow M_v \xrightarrow{p_v} V$$

such that p_v is infinitesimal and which is initial among factorizations of v whose second morphism is infinitesimal.

More explicitly, with $G = P^{\mathrm{gp}}$, put

$$C_v = \left\{ a \in G : \begin{array}{l} \text{there exist } y \in P,\ c \in \mathbb{N} \text{ such that} \\ a \prec y, \quad v(a) \ge 0, \quad v(y) \le c\, v(a) \end{array} \right\}. \tag{2}$$

Then

$$M_v = C_v / (C_v \cap (-C_v)).$$

Moreover,

$$M_v = \operatorname{colim}_{\substack{S \subset C_v \\ S \text{ finite}}} P\langle S \rangle \tag{3}$$

as monoids under P.

> **Reviewer 1**
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> This construction was not part of the authors' solution, but was used by the authors in the larger project from which this problem was drawn.

Proof. For a nonnegative element y, the condition $a \prec y$ can equivalently be written

$$-ky \le a \le ky$$

for some $k \in \mathbb{N}$. Also, for nonnegative b, d, the condition $b \prec d$ is equivalent to $b \le cd$ for some $c \in \mathbb{N}$.

The set C_v contains P. It is closed under addition: if a, a' have witnesses y, y', then $a+a' \prec y+y'$, and, after increasing the two integer bounds to a common bound, $v(y+y')$ is bounded by a multiple of $v(a+a')$.

It is saturated. Suppose $na \in C_v$, with witness y. Choose k such that $ky \pm na \in P$. Since

$$n(ky \pm a) = (ky \pm na) + (n-1)ky \in P,$$

saturation of P gives $ky \pm a \in P$. Saturation of V gives $v(a) \ge 0$, and the inequality bounding $v(y)$ by a multiple of $nv(a)$ is also a bound by a multiple of $v(a)$.

Let $E = v^{-1}(0)$. We claim that

$$C_v \cap (-C_v) = E^{\mathrm{gp}}. \tag{4}$$

If $a, -a \in C_v$, then $v(a) = 0$. A witness y for a therefore satisfies $v(y) = 0$. For sufficiently large k, $a + ky \in P$, and it too has image zero. Thus $a = (a+ky) - ky \in E^{\mathrm{gp}}$. Conversely, a difference of two elements of E is bounded in absolute value by their sum and satisfies (2) in both signs. This proves (4). Sharpening C_v gives a monoid M_v and a well-defined map $p_v : M_v \to V$.

We next verify that p_v is infinitesimal. Let $a \in M_v^{\mathrm{gp}}$ and $b \in M_v$ satisfy

$$p_v(a) \ge 0, \qquad a \prec b, \qquad p_v(b) \prec p_v(a).$$

Choose representatives $x \in G$ and $z \in C_v$ of a, b. A lower bound in $a \prec b$ gives an integer m such that

$$d = x - mz \in C_v.$$

Here an equality modulo the unit subgroup can be lifted in this way because adding or subtracting a unit preserves C_v.

Choose witnesses $y_d, y_z \in P$ for d, z. The inequality $p_v(b) \prec p_v(a)$ gives $v(z) \le e\, v(x)$ for some e. Since $v(d) = v(x) - mv(z) \ge 0$, both $v(d)$ and $v(z)$ are bounded by multiples of $v(x)$. Hence $v(y_d + y_z)$ is bounded by a multiple of $v(x)$. On the other hand,

$$x = d + mz \prec y_d + y_z$$

in the original ordered group P^{gp}. Thus $x \in C_v$, so $a \in M_v$.

Now consider a factorization

$$P \xrightarrow{h} H \xrightarrow{p} V$$

with p infinitesimal. For $a \in C_v$, a witness y in (2) gives

$$h(a) \prec h(y), \qquad p(h(a)) = v(a) \geq 0, \qquad p(h(y)) \prec p(h(a)).$$

Infinitesimality implies $h(a) \in H$. Thus h^{gp} maps C_v into H, kills its units, and induces $M_v \to H$. This map is unique because $P^{\mathrm{gp}} \to M_v^{\mathrm{gp}}$ is surjective. This proves initiality.

Finally, a homomorphism from the right side of (3) to a monoid H is exactly a homomorphism $P \to H$ carrying every element of C_v into H. By the universal property of sharpening, this is exactly a homomorphism $M_v \to H$. This proves the colimit assertion. □

Corollary 5.2. *If $S \subset C_v$ is finite, then*

$$P \longrightarrow P\langle S \rangle$$

is birhizomic at its induced map to V.

Proof. Finite presentation follows from Lemma 4.3. For any infinitesimal factorization of v, the proof of Lemma 5.1 shows that every element of S has nonnegative image in the intermediate monoid. Thus the required lift exists by Lemma 4.3(i). It is unique by that lemma, and its composite to V is the specified map, again by uniqueness. □

5.2 Compactness of valuative tests

Lemma 5.3. *Let $G = P^{\mathrm{gp}}$. The set*

$$K_P = \{W \subset G : P \subset W,\ W + W \subset W,\ W \cup (-W) = G\}$$

is a compact Hausdorff space with the topology induced from $\{0,1\}^G$.

Every $W \in K_P$ defines a valuative monoid

$$V_W = W/(W \cap (-W))$$

and a homomorphism $v_W : P \to V_W$. Conversely, every homomorphism from P to a valuative monoid defines a point of K_P. A finite collection of inequalities between group elements defines a clopen subset of K_P.

Proof. All defining conditions on W are closed conditions involving finitely many coordinates at a time. Thus K_P is closed in the compact Hausdorff product $\{0,1\}^G$.

For completeness, compactness of this product follows directly from the ultrafilter argument. Extend a family of closed subsets with the finite intersection property to an ultrafilter, using Zorn's lemma. At each coordinate the ultrafilter chooses exactly one of the two values. Every basic neighborhood of the resulting point belongs to the ultrafilter, so that point belongs to every closed set in the original family.

The submonoid W is saturated. Indeed, if $ng \in W$ and $g \notin W$, then $-g \in W$, whence

$$g = ng + (n-1)(-g) \in W,$$

a contradiction. Quotienting by its units therefore produces a valuative monoid. Conversely, for $v : P \to V$, take

$$W = \{g \in G : v(g) \geq 0\}.$$

An inequality $v(a) \geq v(b)$ is the coordinate condition $a - b \in W$, proving the last assertion. □

Proposition 5.4 (Finite rational refinements)**.** *Every birhizomic cover of a monoid admits a refinement which is a finite birhizomic cover by finite rational extensions.*

These finite covers form a basis of covering families, stable under base change and composition.

Proof. Let $\{P \to Q_i\}_{i\in I}$ be a birhizomic cover. Fix $W \in K_P$. Apply the covering condition to $v_W : P \to V_W$, and choose

$$P \xrightarrow{f_i} Q_i \xrightarrow{g_i} V_W$$

with f_i birhizomic at g_i. The infinitesimal map $M_{v_W} \to V_W$ gives a lift

$$Q_i \longrightarrow M_{v_W}.$$

Since f_i is of finite presentation, (3) implies that this lift factors through $P\langle S_W\rangle$ for some finite $S_W \subset C_{v_W}$. We obtain a refinement morphism

$$Q_i \longrightarrow P\langle S_W\rangle.$$

For each $s \in S_W$, choose a witness $y_s \in P$, an integer $c_s \geq 0$, and bounds $s \prec y_s$, as in (2). Let

$$O_W = \left\{W' \in K_P : s \in W',\ c_s s - y_s \in W' \text{ for all } s \in S_W\right\}.$$

This is a clopen neighborhood of W. For any valuative homomorphism $v : P \to V$ whose positive cone belongs to O_W, the same witnesses show $S_W \subset C_v$. Corollary 5.2 then shows that $P \to P\langle S_W\rangle$ is birhizomic at its induced map to V.

Compactness gives finitely many such neighborhoods covering K_P. The corresponding rational extensions form a birhizomic cover, by the preceding observation for arbitrary valuative targets, and refine the original cover.

Stability under base change and composition follows from Lemma 4.3 and the topology axioms assumed in the question. □

6 The left Kan extension and its transitivity

Lemma 6.1. *For $F : B_P \to \mathrm{Ab}$, its presheaf left Kan extension to A_P is given by* (1)*. Its indexing categories are filtered, and its restriction to B_P is canonically F.*

Proof. By the universal property of P_x, a morphism $P_x \to Q$ over P exists exactly when $f(x) = 0$, and then it is unique. This gives the usual comma-category formula for left Kan extension.

If x, y are killed by f, then $x + y$ is killed by f, and there are morphisms

$$P_x \longrightarrow P_{x+y}, \qquad P_y \longrightarrow P_{x+y}.$$

There is also the object $P_0 = P$, and all morphisms in the indexing category are unique when they exist. Thus the category is filtered.

For $Q = P_x$, the comma category has the terminal object $P_x \xrightarrow{\mathrm{id}} P_x$, proving the restriction assertion. □

We use the notation $\mathcal{L}_P F$ for this presheaf left Kan extension.

Lemma 6.2 (Transitivity)**.** *Fix $f : P \to Q$, and define a presheaf H on B_Q by*

$$H(Q_y) = (\mathcal{L}_P F)(P \to Q \to Q_y).$$

For every $Q \to R$, there is a canonical isomorphism

$$(\mathcal{L}_Q H)(R) \simeq (\mathcal{L}_P F)(R).$$

Proof. Expanding the left side gives a filtered colimit over pairs (y, x) such that

$$y \in Q, \quad y \mapsto 0 \text{ in } R, \qquad x \in P, \quad x \mapsto 0 \text{ in } Q_y.$$

Every such x is killed in R, so there is a canonical map to the right side.

Conversely, if $x \in P$ is killed in R, take $y = f(x)$. Then y is killed in R, and x is killed in Q_y. Thus every index on the right occurs on the left. All finite collections of these indices and their comparisons can be accommodated at a common larger pair: replace the x's and y's by sums, and add the image of the chosen larger x to the larger y. Hence any equality in the colimit on the right already holds at a common index on the left. This proves that the canonical map is an isomorphism. □

7 A finite convex-cover calculation

7.1 A nerve lemma in the required form

> **Reviewer 1**
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> This reduction to the cohomology of the simplex is the main departure from the human approach. It is a natural solution to the problem, perhaps more 'human' than the human-written version.

Lemma 7.1. *Let U be a nonempty convex subset of a finite-dimensional real vector space. Let $D_1, \dots, D_m$ be closed convex polyhedra such that*

$$U = \bigcup_{i=1}^{m} (U \cap D_i).$$

Let N be the nerve whose nonempty simplices are the subsets $I \subset \{1, \dots, m\}$ for which

$$U \cap \bigcap_{i \in I} D_i \neq \varnothing.$$

For every abelian group A, the augmented simplicial cochain complex of N with coefficients A is exact.

> **Reviewer 1**
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> This is essentially the same simplicial complex used in the authors' solution, but this is presented more concretely.

Proof. For each simplex I of N, choose a point $u_I \in U \cap \bigcap_{i \in I} D_i$, and let K be the convex hull of these finitely many points. Then $K \subset U$ is a nonempty compact convex polytope. The polytopes $K_i = K \cap D_i$ cover K and have exactly the nerve N.

There is a finite triangulation of K in which every K_i is a subcomplex. To construct one, cut K by the finitely many hyperplanes defining the D_i, take all resulting polyhedral cells and their faces, and barycentrically subdivide this finite polyhedral complex.

For nonempty I, put $K_I = \bigcap_{i \in I} K_i$. Consider the double complex

$$D_{p,q} = \bigoplus_{i_0 < \cdots < i_p} C_q(K_{\{i_0, \dots, i_p\}}; \mathbb{Z}),$$

with simplicial boundary vertically and the alternating intersection maps horizontally. The horizontal complex augmented to $C_q(K; \mathbb{Z})$ is exact: for each simplex of the triangulation, its copies are

indexed by the nonempty subsets of the nonempty set of i's whose K_i contain that simplex. This is the augmented chain complex of a simplex.

Every nonempty K_I is convex and hence contractible. Consequently its simplicial homology is $\mathbb{Z}$ in degree zero and zero in higher degrees. Here one may use homotopy invariance and the simplicial–singular comparison [1, Theorems 2.10 and 2.27]. Computing the homology of the finite double complex in its two directions shows that the nerve has the homology of K. The double-complex argument is the standard one recalled in [2, Tag 012X]. Thus the augmented integral chain complex of N is exact.

This is a finite exact complex of finitely generated free abelian groups, including its augmentation to $\mathbb{Z}$. Its cycle groups are free, so its short exact sequences split. Applying $\mathrm{Hom}_{\mathbb{Z}}(-, A)$ therefore preserves exactness and gives the claimed augmented cochain complex. □

7.2 Weak valuative covers

Definition 7.2. A finite family

$$T \longrightarrow T\langle S_i\rangle, \qquad 1 \le i \le m,$$

is *weakly valuative* if every homomorphism from T to a valuative monoid factors through at least one member of the family. No infinitesimal lifting condition is imposed in this definition.

Every birhizomic cover is weakly valuative. For a nonempty subset $I \subset \{1, \dots, m\}$, write

$$T_I = T\left\langle \bigcup_{i\in I} S_i \right\rangle.$$

These are the intersections in the opposite site.

Proposition 7.3 (The fs Čech calculation)**.** *Suppose T is fs, $H : B_T \to \mathrm{Ab}$ is arbitrary, and $\{T \to T\langle S_i\rangle\}_{i=1}^m$ is weakly valuative. Then the augmented alternating complex*

$$0 \longrightarrow H(T) \longrightarrow \prod_i (\mathcal{L}_T H)(T_i) \longrightarrow \prod_{i<j} (\mathcal{L}_T H)(T_{\{i,j\}}) \longrightarrow \cdots \longrightarrow (\mathcal{L}_T H)(T_{\{1,\dots,m\}}) \longrightarrow 0 \tag{5}$$

is exact.

> **Reviewer 1**
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> This is the key finite-type calculation.

Proof. By Lemma 4.5, we may identify B_T with the finite poset of faces of T. For a face E, let T_E denote T_x for any x with $E_x = E$. If R is a monoid under T, its kernel $E_R = \ker(T \to R)$ is a face, so

$$(\mathcal{L}_T H)(R) = H(T_{E_R}). \tag{6}$$

Set $L = T^{\mathrm{gp}}$, and consider the polyhedral cones

$$\sigma = \{\ell \in \mathrm{Hom}(L, \mathbb{R}) : \ell(T) \ge 0\}, \qquad \sigma_i = \{\ell \in \sigma : \ell(s) \ge 0 \text{ for all } s \in S_i\}.$$

Weak valuative covering, applied to $\mathbb{R}_{\ge 0}$, implies

$$\sigma = \bigcup_i \sigma_i.$$

For nonempty I, put $\sigma_I = \bigcap_{i\in I} \sigma_i$ and $E_I = \ker(T \to T_I)$. The universal property of rational extensions identifies σ_I with the homomorphisms $T_I \to \mathbb{R}_{\geq 0}$. Lemma 4.6 gives

$$E_I = \{a \in T : \ell(a) = 0 \text{ for every } \ell \in \sigma_I\}. \tag{7}$$

Indeed, a nonzero element of the fs monoid T_I is detected by a strictly positive real homomorphism.

Fix a face E of T and an abelian group A. Define a functor $J_E(A)$ on the face poset by

$$J_E(A)(T_D) = \begin{cases} A, & D \subset E, \\ 0, & D \not\subset E, \end{cases}$$

with identity transition maps whenever both values are A.

Choose generators $p_1, \ldots, p_r$ of T, and put

$$U_E = \{\ell \in \sigma : \ell(p_j) > 0 \text{ for every } p_j \notin E\}.$$

This is nonempty and convex: nonemptiness follows from the strictly positive functional in Lemma 4.6, with the case $T = \{0\}$ interpreted in the evident way.

We claim that

$$E_I \subset E \quad \Longleftrightarrow \quad \sigma_I \cap U_E \neq \varnothing. \tag{8}$$

For the forward implication, each generator $p_j \notin E$ is outside E_I. By (7), there is an element of σ_I positive on that generator. Summing these finitely many functionals gives an element of $\sigma_I \cap U_E$. If there are no such generators, the zero functional suffices.

Conversely, if $E_I \not\subset E$, express an element of $E_I \setminus E$ in the chosen generators. Since E_I is a face, every generator occurring belongs to E_I; since E is a face, at least one occurring generator lies outside E. Every functional in σ_I vanishes on that generator, so $\sigma_I \cap U_E = \varnothing$.

For $H = J_E(A)$, equations (6) and (8) identify (5) with the augmented cochain complex of the nerve of

$$U_E = \bigcup_i (U_E \cap \sigma_i).$$

It is exact by Lemma 7.1.

It remains to pass from these functors to arbitrary H. The functor J_E is right adjoint to evaluation at T_E:

$$\operatorname{Hom}(H, J_E(A)) = \operatorname{Hom}(H(T_E), A).$$

Thus $J_E(A)$ is injective when A is an injective abelian group. There are enough such injectives: choose embeddings $H(T_E) \hookrightarrow A_E$ into injective abelian groups and use the adjunctions to obtain

$$H \hookrightarrow \prod_E J_E(A_E).$$

The component indexed by $E = D$ proves injectivity at T_D. The existence of injective embeddings of abelian groups is standard; see [2, Tag 01D6]. Because the face poset is finite, every injective functor is a direct summand of a finite product of functors of this form. Consequently (5) is exact for injective H.

Finally, all terms of (5), including its augmentation, are exact functors of H, by (6). Apply them to an injective resolution of H. The resulting double complex has exact augmented rows, while its vertical cohomology is the augmented complex for H. Its horizontal direction is bounded, and, after shifting the augmented degree by one, it is a first-quadrant double complex. Computing total cohomology in the two directions proves exactness for arbitrary H; see again [2, Tag 012X]. □

> **Reviewer 1**
>
> The proof is correct. It combines several ideas that might better be split into separate statements.

8 Passage to arbitrary monoids

Lemma 8.1. *Let $T = \bigcup_\lambda T_\lambda$ be a filtered union of fs submonoids. Suppose finite sets $S_1, \dots, S_m \subset T^{\mathrm{gp}}$ define a weakly valuative family over T. After restricting to a cofinal set of indices, all S_i lie in T_λ^{gp}, and*

$$T_\lambda \longrightarrow T_\lambda\langle S_i\rangle, \qquad 1 \le i \le m,$$

is weakly valuative for every remaining λ.

Proof. The finite sets S_i belong to a common T_λ^{gp}, so we first restrict to such indices.

The compact spaces of Lemma 5.3 satisfy

$$K_T \simeq \varprojlim_\lambda K_{T_\lambda}.$$

Indeed, the groups T_λ^{gp} have union T^{gp}, and compatible positive cones on them give exactly a positive cone containing T.

In K_{T_λ}, put

$$D_{i,\lambda} = \{W : S_i \subset W\}, \qquad Z_\lambda = K_{T_\lambda} \setminus \bigcup_i D_{i,\lambda}.$$

These are clopen sets, and the Z_λ form an inverse system of compact spaces. If every Z_λ were nonempty, their inverse limit would be nonempty: every finite collection of compatibility requirements can be satisfied by choosing a point at a common upper index, and compactness then supplies a compatible family. Such a family would define a point of K_T lying in none of the D_i, contrary to weak valuative covering over T.

Thus $Z_\lambda = \varnothing$ at some stage and at every later stage. By Lemma 4.3(i), this is precisely the desired weak valuative covering condition. □

Proposition 8.2 (The general Čech calculation)**.** *Proposition 7.3 holds for an arbitrary monoid T.*

> **Reviewer 1**
>
> This is done the same way as in the human solution, but is more explicit.

Proof. Choose a filtered union $T = \bigcup_\lambda T_\lambda$ as in Lemma 4.6. By Lemma 8.1, we may suppose that the given finite rational family is defined and weakly valuative at every stage.

For $x \in T_\lambda$, base change gives

$$(T_\lambda)_x \amalg_{T_\lambda} T \simeq T_x.$$

Define $H_\lambda : B_{T_\lambda} \to \mathrm{Ab}$ by

$$H_\lambda((T_\lambda)_x) = H(T_x).$$

For nonempty $I \subset \{1, \dots, m\}$, write

$$T_{I,\lambda} = T_\lambda\left\langle \bigcup_{i\in I} S_i \right\rangle, \qquad T_I = T\left\langle \bigcup_{i\in I} S_i \right\rangle.$$

The universal property of rational extensions gives

$$T_I = \mathrm{colim}_\lambda\, T_{I,\lambda}.$$

Let $E_{I,\lambda} = \ker(T_\lambda \to T_{I,\lambda})$, and choose $x_{I,\lambda} \in T_\lambda$ generating this face in the sense of Lemma 4.5. Then

$$(\mathcal{L}_{T_\lambda} H_\lambda)(T_{I,\lambda}) = H(T_{x_{I,\lambda}}).$$

There is a natural isomorphism

$$(\mathcal{L}_T H)(T_I) \simeq \operatorname{colim}_\lambda (\mathcal{L}_{T_\lambda} H_\lambda)(T_{I,\lambda}). \tag{9}$$

To verify it, an element $a \in T$ killed in T_I belongs to some T_λ and becomes zero in a later $T_{I,\mu}$, by Lemma 4.1. It therefore belongs to $E_{I,\mu}$, so $a \le n x_{I,\mu}$ in T_μ for some n. Hence the objects $T_{x_{I,\lambda}}$ are cofinal in the indexing category defining $(\mathcal{L}_T H)(T_I)$. The same reasoning gives the compatibility maps and proves (9).

At the augmentation, $H_\lambda(T_\lambda) = H(T)$. There are only finitely many terms in each degree of the alternating Čech complex. Therefore (9) identifies the augmented complex over T with the filtered colimit of the augmented complexes over T_λ. The latter are exact by Proposition 7.3. Filtered colimits of abelian groups are exact, and they commute with finite products, so the complex over T is exact as well; for exactness of filtered colimits, see [2, Tag 00DB]. □

Remark 8.3. Only weak valuative covering was descended in this argument. There is no need to descend the infinitesimal lifting condition, and there is no finite-presentation assumption on the coefficient presheaf.

9 From Čech exactness to acyclicity

We include the sheaf-theoretic argument to make explicit why the preceding calculations imply vanishing of derived, rather than merely Čech, cohomology.

Lemma 9.1. *Let $\mathcal{C}$ be a site admitting a basis of finite covering families by monomorphisms, stable under base change and composition, with the corresponding finite intersections. Suppose an abelian presheaf $\mathcal{E}$ has exact augmented alternating Čech complex for every covering family in this basis. Then $\mathcal{E}$ is a sheaf and*

$$H^q(U, \mathcal{E}) = 0 \qquad \text{for every } U \in \mathcal{C},\ q > 0.$$

Proof. Exactness in augmented degrees $-1, 0$ is the sheaf condition for the basis. It implies the sheaf condition for all covers: refine a cover by a basis cover to glue, and use base changes of that basis cover to verify that the glued section restricts to the originally specified sections.

We first recall why an injective abelian sheaf $\mathcal{I}$ has exact augmented alternating Čech complex for a finite cover by monomorphisms. Let $\mathbb{Z}[U]$ denote the free abelian sheaf on the associated representable sheaf of U. The augmented chain complex

$$\cdots \longrightarrow \bigoplus_{i<j} \mathbb{Z}[U_i \times_U U_j] \longrightarrow \bigoplus_i \mathbb{Z}[U_i] \longrightarrow \mathbb{Z}[U] \longrightarrow 0$$

is exact. Locally over a section of U, one of the covering subobjects is available, and inserting its index contracts the augmented complex. This local exactness is exactness after sheafification. Applying the exact contravariant functor $\operatorname{Hom}(-, \mathcal{I})$, and using $\operatorname{Hom}(\mathbb{Z}[U], \mathcal{I}) = \mathcal{I}(U)$, proves the assertion. This is also the usual injective Čech-acyclicity statement; see [2, Tag 03AW].

Embed $\mathcal{E}$ in an injective abelian sheaf $\mathcal{I}$, and let $\mathcal{Q} = \mathcal{I}/\mathcal{E}$. Such embeddings exist on any site [2, Tag 01DP]. We claim that

$$0 \longrightarrow \mathcal{E}(U) \longrightarrow \mathcal{I}(U) \longrightarrow \mathcal{Q}(U) \longrightarrow 0 \tag{10}$$

is exact for every object U.

Indeed, a section of $\mathcal{Q}(U)$ lifts locally to $\mathcal{I}$, and we may choose a finite basis cover for these lifts. Local lifting for epimorphisms of abelian sheaves is part of the usual description of sheaf cokernels [2, Tag 03CM]. The differences of the local lifts form a Čech 1-cocycle with values in $\mathcal{E}$. By the assumed exactness, this is a coboundary. Adjusting the lifts by a 0-cochain in $\mathcal{E}$ makes them agree, and the sheaf condition for $\mathcal{I}$ glues them to a global lift. This proves (10).

For every basis cover, (10) yields a short exact sequence of augmented Čech complexes. The complexes for $\mathcal{E}$ and $\mathcal{I}$ are exact, so the complex for $\mathcal{Q}$ is exact. Thus $\mathcal{Q}$ has the same property as $\mathcal{E}$.

The long exact sequence of the right derived functors of evaluation, applied to $0 \to \mathcal{E} \to \mathcal{I} \to \mathcal{Q} \to 0$, gives

$$H^1(U, \mathcal{E}) = 0, \qquad H^q(U, \mathcal{E}) \simeq H^{q-1}(U, \mathcal{Q}) \quad (q \geq 2).$$

Repeating the argument with $\mathcal{Q}$ proves vanishing in every positive degree. □

> **Reviewer 1**
>
> This is a nice proof of a standard fact, but could have been omitted.

10 Proof of the theorem

Proof of Theorem 3.1. Let

$$\mathcal{G}_F = \mathcal{L}_P F.$$

Lemma 6.1 proves formula (1) and identifies its restriction to B_P with F.

Consider any object $P \to Q$. By Proposition 5.4, every birhizomic cover of this object has a refinement of the form

$$Q \longrightarrow Q\langle S_i \rangle, \qquad 1 \leq i \leq m,$$

which is a finite birhizomic cover. These finite rational covers form a basis, and their opposites are monomorphisms by Remark 4.4.

Define $H : B_Q \to \mathrm{Ab}$ by

$$H(Q_y) = \mathcal{G}_F(P \to Q \to Q_y).$$

Lemma 6.2 identifies $\mathcal{G}_F$, on monoids under Q, with $\mathcal{L}_Q H$. The displayed covering family is weakly valuative. Consequently, Proposition 8.2 proves exactness of its augmented alternating Čech complex with coefficients in $\mathcal{G}_F$.

This applies to every object Q and every basis cover. Lemma 9.1 therefore gives both

$$\mathcal{G}_F \in \mathrm{Sh}(A_P^{\mathrm{op}}, \mathrm{Ab}) \quad \text{and} \quad H^q(Q, \mathcal{G}_F) = 0 \quad (q > 0).$$

Since $\mathcal{G}_F$ is the presheaf left Kan extension and is already a sheaf, it is canonically $\pi^{-1}F$. This proves (i) and (ii).

Finally, the right adjoint π_* is restriction to B_P. Indeed, restriction is right adjoint to presheaf left Kan extension, and sheafification is left adjoint to inclusion of sheaves into presheaves. The topology on B_P^{op} is trivial, so no additional sheafification is involved on that side.

Take an injective resolution $\mathcal{G}_F \to \mathcal{I}^\bullet$ in abelian sheaves on A_P^{op}. Because cohomology in the presheaf category on B_P is computed objectwise, for every $x \in P$ we have

$$\big(\mathrm{R}^q \pi_* \mathcal{G}_F\big)(P_x) = H^q\big(\mathcal{I}^\bullet(P_x)\big) = H^q(P_x, \mathcal{G}_F).$$

These groups vanish for $q > 0$. In degree zero, the canonical left Kan extension unit gives

$$(\pi_* \mathcal{G}_F)(P_x) = \mathcal{G}_F(P_x) = F(P_x).$$

Thus the derived unit induces the identity isomorphism in degree zero and has zero higher cohomology on its target. It is therefore a quasi-isomorphism:

$$\boxed{F \xrightarrow{\sim} \mathrm{R}\pi_* \pi^{-1} F}.$$

□

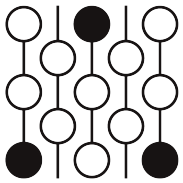
References

[1] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Theorems 2.10 and 2.27.

[2] The Stacks Project Authors, *The Stacks Project*. Tags `00DB` (exactness of filtered colimits), `012X` (spectral sequences of double complexes), `01D6` (injective abelian groups), `01DP` (enough injectives for abelian sheaves on a site), `03CM` (abelian sheaves and local exactness), and `03AW` (injective Čech acyclicity).

Review of Problem 1 – Astra

First Proof Third Batch
Reviewer 2
October 8–9, 2026



1 Review

This submission included a complete proof that was well-written and thorough. Below we highlight some of the features characteristic to the solution and style of the submission, but the proof and writing were essentially perfect.

[Identifying passage redacted.]

The proof reduces to a finite-dimensional approximation of a general cover, as well as noting that the "weakly valuative" topology suffices here for cohomological vanishing. Its Cech computation reduced to a more topological argument for contractibility of the nerve in Lemma 7.1, citing Hatcher. After showing Cech cohomology vanishes, it results that sheaf cohomology does.

Compared to other submissions, it included some interesting material on the rhizomic topology. For example, in Lemma 5.1, it describes a universal (formally) birhizomic/infinitesimal factorization. On the other hand, it included Lemma 9.1 which is a standard argument which could have been a citation. In a couple of instances, it refrained from identifying a certain construction with its typical name or property in the literature: the extension by zero is not identified by name or universal property, but the latter is used in Proposition 7.3, and it spells out a stricter notion of exactness of every section rather than using the phrase "exactness as presheaves." It was ever so slightly cavalier with finiteness hypotheses in Proposition 8.2, saying that a filtered union was "as in Lemma 4.6" instead of identifying the fs properties of the T_λ explicitly which it needed. These small turns of phrase amount to stylistic points, as the proof is excellent, thorough, and clear.

We struggled to agree on a comparison between this submission and the Heron submission; they are approximately equal in quality.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 1

2 Submission with inline comments

Abstract

The answer to the question is affirmative, without a finite-generation hypothesis on the monoid. We prove a stronger statement: the presheaf left Kan extension of the given presheaf is already a birhizomic sheaf and is acyclic on every object of the birhizomic site. The proof uses finite refinements of birhizomic covers, a finite convex-cover calculation, and passage to filtered unions of finitely generated saturated monoids.

3 Statement and conventions

We use the definition of monoid in the question. In intermediate constructions, submonoids of groups are allowed to have units; we explicitly sharpen them before regarding them as objects of the category in the question. All group-valued expressions involving a monoid homomorphism refer to its extension to groupifications.

Put

$$P_x = P[-x]^\sharp \qquad (x \in P).$$

A presheaf on B_P^{op} will be regarded as a covariant functor $B_P \to \mathrm{Ab}$. Likewise, presheaves on A_P^{op} are covariant functors on A_P.

For a sheaf $\mathcal{E}$ on A_P^{op}, write $H^q(Q, \mathcal{E})$ for the right derived functors of evaluation at the object $P \to Q$. We make the usual universe choices for sites and sheaf categories.

Theorem 3.1. *Let P be any monoid, and let $F : B_P \to \mathrm{Ab}$ be any functor. For $f : P \to Q$, define*

$$\mathcal{G}_F(Q) = \operatorname{colim}_{\substack{x \in P \\ f(x)=0}} F(P_x), \tag{1}$$

where the indexing category consists of the corresponding objects P_x and their morphisms over P. Then:

(i) *$\mathcal{G}_F$ is the presheaf left Kan extension of F, and it is already a sheaf for the birhizomic topology;*

(ii) *for every $P \to Q$,*

$$H^q(Q, \mathcal{G}_F) = 0 \qquad (q > 0);$$

(iii) *the canonical unit is a quasi-isomorphism:*

$$F \xrightarrow{\sim} \mathrm{R}\pi_* \pi^{-1} F.$$

The indexing category in (1) is filtered, as will be proved below. In particular, the formula does not require its indexing face to be finitely generated.

4 Elementary constructions with monoids

4.1 Filtered colimits and finite rational extensions

Lemma 4.1. *The groupification of a monoid is torsion-free. Filtered colimits of monoids are computed on the underlying commutative monoids, and hence on the underlying sets.*

Proof. If $ng = 0$ in P^{gp}, with $n > 0$, saturation gives $g, -g \in P$, and sharpness gives $g = 0$.

Let $P = \operatorname{colim}_\lambda P_\lambda$ be a filtered colimit of the underlying commutative monoids. Integrality and sharpness hold in P: any equality witnessing a failure of either property holds at some stage, where that property is valid.

To check saturation, write an element of P^{gp} as $a-b$, with $a, b \in P$, and suppose $n(a-b) = c \in P$. By integrality, this is the equality $na = nb + c$ in P. Representatives for a, b, c, together with this equality, exist at a common stage. Saturation at that stage shows that $a - b$ belongs to its positive monoid, and therefore belongs to P. □

Definition 4.2. For a finite subset $S \subset P^{\mathrm{gp}}$, define

$$C_P(S) = \left\{ g \in P^{\mathrm{gp}} : ng \in P + \sum_{s \in S} \mathbb{N}s \text{ for some } n > 0 \right\},$$

and put

$$P\langle S\rangle = C_P(S)/\big(C_P(S) \cap (-C_P(S))\big).$$

We call $P \to P\langle S\rangle$ a *finite rational extension*. This terminology is used only for the construction just specified.

Lemma 4.3. *The construction in Definition 4.2 has the following properties.*

(i) *For any $h : P \to M$, there is a morphism $P\langle S\rangle \to M$ extending h if and only if $h(s) \in M$ for all $s \in S$. Such a morphism is unique.*

(ii) *The morphism $P \to P\langle S\rangle$ is of finite presentation, and its groupification is surjective.*

(iii) *Finite rational extensions are preserved by arbitrary pushout and by composition. Moreover,*

$$P\langle S\rangle \amalg_P P\langle T\rangle \simeq P\langle S \cup T\rangle.$$

(iv) *$P\langle\{-x\}\rangle \simeq P_x$ for $x \in P$.*

Proof. The submonoid $C = C_P(S)$ is saturated in P^{gp}, and its units form the subgroup $U = C \cap (-C)$. The quotient C/U is saturated: if $n(g + U)$ belongs to C/U, then $ng \in C$, since adding or subtracting an element of U does not change membership in C. Thus $g \in C$. The quotient is integral and sharp by construction.

If $h(s) \in M$ for all $s \in S$, saturation of M implies that $h(C) \subset M$, and sharpness implies that $h(U) = 0$. This gives the asserted extension. Its uniqueness follows because $P^{\mathrm{gp}} \to (P\langle S\rangle)^{\mathrm{gp}}$ is surjective.

Write $s = a_s - b_s$, with $a_s, b_s \in P$. The condition $h(s) \in M$ is equivalent to the existence of an element $z_s \in M$ such that

$$h(b_s) + z_s = h(a_s).$$

There are finitely many such conditions, and each z_s, when it exists, is unique by integrality. Lemma 4.1 therefore shows that $\mathrm{Hom}_P(P\langle S\rangle, -)$ commutes with filtered colimits. This proves finite presentation.

The pushout assertion follows immediately from (i). More generally, under $h : P \to Q$, the pushout is $Q\langle h(S)\rangle$. For composition, lift the finitely many group elements used in a rational extension of $P\langle S\rangle$ to P^{gp}; adjoining these lifts together with S gives the composite.

Finally, $h(-x) \in M$, with $h(x) \in M$, is equivalent to $h(x) = 0$, since M is sharp. This is precisely the universal property of P_x. □

Remark 4.4. Every finite rational extension is an epimorphism in the category of monoids, by the uniqueness assertion in Lemma 4.3(i). Consequently, its opposite is a monomorphism of the site. This observation will allow us to use finite alternating Čech complexes.

4.2 Faces and the category B_P

A *face* of P is a submonoid $E \subset P$ such that $a + b \in E$ implies $a, b \in E$. The inverse image of 0 under a homomorphism of sharp monoids is a face.

Lemma 4.5. *For $x \in P$, set*

$$E_x = \{a \in P : a \leq nx \text{ for some } n \in \mathbb{N}\}.$$

Then $E_x = \ker(P \to P_x)$, and

$$\mathrm{Hom}_P(P_x, P_y) \neq \varnothing \quad \Longleftrightarrow \quad E_x \subset E_y.$$

When this set of morphisms is nonempty, it is a singleton.

If P is finitely generated, every face is of the form E_x, there are only finitely many faces, and B_P, up to equivalence, is their inclusion poset.

Proof. The localization $P + \mathbb{N}(-x) \subset P^{\mathrm{gp}}$ is already saturated. Indeed, if $ng = a - kx$, with $a \in P$ and $k \geq 0$, then

$$n(g + kx) = a + (n-1)kx \in P,$$

so $g + kx \in P$. An element $a \in P$ becomes a unit in this localization precisely when $a \leq nx$ for some n. This proves the kernel formula. The assertion about morphisms follows from the universal property of P_x.

If $p_1, \ldots, p_r$ generate P, a face E is generated by those p_j which belong to E: every summand in an expression for an element of a face belongs to that face. Put

$$x_E = \sum_{p_j \in E} p_j.$$

Every element of E is bounded above by a multiple of x_E, and the face property gives $E_{x_E} \subset E$. Hence $E = E_{x_E}$. The same description proves finiteness of the set of faces. □

4.3 Finite-dimensional approximation

We call a monoid *fs* if it is finitely generated, in addition to the standing assumptions.

Lemma 4.6. *The following assertions hold.*

(i) *Every monoid is a filtered union of fs submonoids.*

(ii) *If P is fs and $S \subset P^{\mathrm{gp}}$ is finite, then $P\langle S\rangle$ is fs.*

(iii) *If P is fs, there is a homomorphism $\ell : P \to \mathbb{R}_{\geq 0}$ which is strictly positive on $P \setminus \{0\}$.*

Proof. We first recall the relevant elementary lattice calculation. Let L be a finitely generated free abelian group, and let $v_1, \ldots, v_r \in L$. If

$$C = \sum_j \mathbb{R}_{\geq 0} v_j,$$

then the saturation in L of $\sum_j \mathbb{N} v_j$ is $C \cap L$. Indeed, a nonnegative real solution to a linear system with rational coefficients and rational right-hand side can be replaced by a nonnegative rational solution. One way to see this is to choose a solution with minimal support: dependence among its supporting columns permits movement until a coordinate becomes zero; with independent supporting columns, the solution is rational by linear algebra. Clearing denominators proves the assertion.

Furthermore, $C \cap L$ is finitely generated. For $g = \sum_j t_j v_j \in C \cap L$, subtract $\sum_j \lfloor t_j \rfloor v_j$. The remainder is a lattice point in the bounded set

$$\left\{ \sum_j u_j v_j : 0 \le u_j \le 1 \right\}.$$

There are only finitely many such remainders, and they, together with the v_j, generate $C \cap L$.

For a finite subset of an arbitrary monoid P, take the saturation of the submonoid it generates in the subgroup it generates in P^{gp}. That subgroup is a finitely generated torsion-free abelian group, hence is free. The saturation lies in P, is finitely generated by the preceding calculation, and is sharp. These submonoids are directed under inclusion and exhaust P. This proves (i). Applying the same calculation to generators of P together with S, and then taking the quotient by units, proves (ii).

For (iii), assume $P \neq \{0\}$, and choose nonzero generators $p_1, \ldots, p_r$. The origin does not belong to their convex hull. Otherwise the rational-solution argument above would give a nontrivial relation

$$\sum_j n_j p_j = 0, \qquad n_j \in \mathbb{N},$$

contrary to sharpness. Choose an inner product on $P^{\mathrm{gp}} \otimes \mathbb{R}$, and let z be the point in this compact convex hull nearest the origin. Then $z \neq 0$, and minimality gives

$$\langle z, p_j \rangle \ge \|z\|^2 > 0$$

for every j. The functional $u \mapsto \langle z, u \rangle$ has the required property. For $P = \{0\}$, the assertion is immediate. □

5 Infinitesimal envelopes and finite covering refinements

5.1 An explicit infinitesimal envelope

Lemma 5.1. *Let $v : P \to V$ be a homomorphism to a valuative monoid. There is a factorization*

$$P \longrightarrow M_v \xrightarrow{p_v} V$$

such that p_v is infinitesimal and which is initial among factorizations of v whose second morphism is infinitesimal.

More explicitly, with $G = P^{\mathrm{gp}}$, put

$$C_v = \left\{ a \in G : \begin{array}{l} \text{there exist } y \in P,\ c \in \mathbb{N} \text{ such that} \\ a \prec y, \quad v(a) \ge 0, \quad v(y) \le c\, v(a) \end{array} \right\}. \tag{2}$$

Then

$$M_v = C_v / (C_v \cap (-C_v)).$$

Moreover,

$$M_v = \operatorname{colim}_{\substack{S \subset C_v \\ S \text{ finite}}} P\langle S \rangle \tag{3}$$

as monoids under P.

Proof. For a nonnegative element y, the condition $a \prec y$ can equivalently be written

$$-ky \le a \le ky$$

for some $k \in \mathbb{N}$. Also, for nonnegative b, d, the condition $b \prec d$ is equivalent to $b \le cd$ for some $c \in \mathbb{N}$.

The set C_v contains P. It is closed under addition: if a, a' have witnesses y, y', then $a+a' \prec y+y'$, and, after increasing the two integer bounds to a common bound, $v(y+y')$ is bounded by a multiple of $v(a+a')$.

It is saturated. Suppose $na \in C_v$, with witness y. Choose k such that $ky \pm na \in P$. Since

$$n(ky \pm a) = (ky \pm na) + (n-1)ky \in P,$$

saturation of P gives $ky \pm a \in P$. Saturation of V gives $v(a) \ge 0$, and the inequality bounding $v(y)$ by a multiple of $nv(a)$ is also a bound by a multiple of $v(a)$.

Let $E = v^{-1}(0)$. We claim that

$$C_v \cap (-C_v) = E^{\mathrm{gp}}. \tag{4}$$

If $a, -a \in C_v$, then $v(a) = 0$. A witness y for a therefore satisfies $v(y) = 0$. For sufficiently large k, $a + ky \in P$, and it too has image zero. Thus $a = (a+ky) - ky \in E^{\mathrm{gp}}$. Conversely, a difference of two elements of E is bounded in absolute value by their sum and satisfies (2) in both signs. This proves (4). Sharpening C_v gives a monoid M_v and a well-defined map $p_v : M_v \to V$.

We next verify that p_v is infinitesimal. Let $a \in M_v^{\mathrm{gp}}$ and $b \in M_v$ satisfy

$$p_v(a) \ge 0, \qquad a \prec b, \qquad p_v(b) \prec p_v(a).$$

Choose representatives $x \in G$ and $z \in C_v$ of a, b. A lower bound in $a \prec b$ gives an integer m such that

$$d = x - mz \in C_v.$$

Here an equality modulo the unit subgroup can be lifted in this way because adding or subtracting a unit preserves C_v.

Choose witnesses $y_d, y_z \in P$ for d, z. The inequality $p_v(b) \prec p_v(a)$ gives $v(z) \le e\, v(x)$ for some e. Since $v(d) = v(x) - mv(z) \ge 0$, both $v(d)$ and $v(z)$ are bounded by multiples of $v(x)$. Hence $v(y_d + y_z)$ is bounded by a multiple of $v(x)$. On the other hand,

$$x = d + mz \prec y_d + y_z$$

in the original ordered group P^{gp}. Thus $x \in C_v$, so $a \in M_v$.

Now consider a factorization

$$P \xrightarrow{h} H \xrightarrow{p} V$$

with p infinitesimal. For $a \in C_v$, a witness y in (2) gives

$$h(a) \prec h(y), \qquad p(h(a)) = v(a) \ge 0, \qquad p(h(y)) \prec p(h(a)).$$

Infinitesimality implies $h(a) \in H$. Thus h^{gp} maps C_v into H, kills its units, and induces $M_v \to H$. This map is unique because $P^{\mathrm{gp}} \to M_v^{\mathrm{gp}}$ is surjective. This proves initiality.

Finally, a homomorphism from the right side of (3) to a monoid H is exactly a homomorphism $P \to H$ carrying every element of C_v into H. By the universal property of sharpening, this is exactly a homomorphism $M_v \to H$. This proves the colimit assertion. □

Corollary 5.2. *If $S \subset C_v$ is finite, then*

$$P \longrightarrow P\langle S\rangle$$

is birhizomic at its induced map to V.

Proof. Finite presentation follows from Lemma 4.3. For any infinitesimal factorization of v, the proof of Lemma 5.1 shows that every element of S has nonnegative image in the intermediate monoid. Thus the required lift exists by Lemma 4.3(i). It is unique by that lemma, and its composite to V is the specified map, again by uniqueness. □

5.2 Compactness of valuative tests

Lemma 5.3. *Let $G = P^{\mathrm{gp}}$. The set*

$$K_P = \{W \subset G : P \subset W,\ W + W \subset W,\ W \cup (-W) = G\}$$

is a compact Hausdorff space with the topology induced from $\{0,1\}^G$.

Every $W \in K_P$ defines a valuative monoid

$$V_W = W/(W \cap (-W))$$

and a homomorphism $v_W : P \to V_W$. Conversely, every homomorphism from P to a valuative monoid defines a point of K_P. A finite collection of inequalities between group elements defines a clopen subset of K_P.

Proof. All defining conditions on W are closed conditions involving finitely many coordinates at a time. Thus K_P is closed in the compact Hausdorff product $\{0,1\}^G$.

For completeness, compactness of this product follows directly from the ultrafilter argument. Extend a family of closed subsets with the finite intersection property to an ultrafilter, using Zorn's lemma. At each coordinate the ultrafilter chooses exactly one of the two values. Every basic neighborhood of the resulting point belongs to the ultrafilter, so that point belongs to every closed set in the original family.

The submonoid W is saturated. Indeed, if $ng \in W$ and $g \notin W$, then $-g \in W$, whence

$$g = ng + (n-1)(-g) \in W,$$

a contradiction. Quotienting by its units therefore produces a valuative monoid. Conversely, for $v : P \to V$, take

$$W = \{g \in G : v(g) \geq 0\}.$$

An inequality $v(a) \geq v(b)$ is the coordinate condition $a - b \in W$, proving the last assertion. □

Proposition 5.4 (Finite rational refinements)**.** *Every birhizomic cover of a monoid admits a refinement which is a finite birhizomic cover by finite rational extensions.*

These finite covers form a basis of covering families, stable under base change and composition.

Proof. Let $\{P \to Q_i\}_{i \in I}$ be a birhizomic cover. Fix $W \in K_P$. Apply the covering condition to $v_W : P \to V_W$, and choose

$$P \xrightarrow{f_i} Q_i \xrightarrow{g_i} V_W$$

with f_i birhizomic at g_i. The infinitesimal map $M_{v_W} \to V_W$ gives a lift

$$Q_i \longrightarrow M_{v_W}.$$

Since f_i is of finite presentation, (3) implies that this lift factors through $P\langle S_W\rangle$ for some finite $S_W \subset C_{v_W}$. We obtain a refinement morphism

$$Q_i \longrightarrow P\langle S_W\rangle.$$

For each $s \in S_W$, choose a witness $y_s \in P$, an integer $c_s \geq 0$, and bounds $s \prec y_s$, as in (2). Let

$$O_W = \left\{W' \in K_P : s \in W',\ c_s s - y_s \in W' \text{ for all } s \in S_W\right\}.$$

This is a clopen neighborhood of W. For any valuative homomorphism $v : P \to V$ whose positive cone belongs to O_W, the same witnesses show $S_W \subset C_v$. Corollary 5.2 then shows that $P \to P\langle S_W\rangle$ is birhizomic at its induced map to V.

Compactness gives finitely many such neighborhoods covering K_P. The corresponding rational extensions form a birhizomic cover, by the preceding observation for arbitrary valuative targets, and refine the original cover.

Stability under base change and composition follows from Lemma 4.3 and the topology axioms assumed in the question. □

6 The left Kan extension and its transitivity

Lemma 6.1. *For $F : B_P \to \mathrm{Ab}$, its presheaf left Kan extension to A_P is given by* (1)*. Its indexing categories are filtered, and its restriction to B_P is canonically F.*

Proof. By the universal property of P_x, a morphism $P_x \to Q$ over P exists exactly when $f(x) = 0$, and then it is unique. This gives the usual comma-category formula for left Kan extension.

If x, y are killed by f, then $x + y$ is killed by f, and there are morphisms

$$P_x \longrightarrow P_{x+y}, \qquad P_y \longrightarrow P_{x+y}.$$

There is also the object $P_0 = P$, and all morphisms in the indexing category are unique when they exist. Thus the category is filtered.

For $Q = P_x$, the comma category has the terminal object $P_x \xrightarrow{\mathrm{id}} P_x$, proving the restriction assertion. □

We use the notation $\mathcal{L}_P F$ for this presheaf left Kan extension.

Lemma 6.2 (Transitivity)**.** *Fix $f : P \to Q$, and define a presheaf H on B_Q by*

$$H(Q_y) = (\mathcal{L}_P F)(P \to Q \to Q_y).$$

For every $Q \to R$, there is a canonical isomorphism

$$(\mathcal{L}_Q H)(R) \simeq (\mathcal{L}_P F)(R).$$

Proof. Expanding the left side gives a filtered colimit over pairs (y, x) such that

$$y \in Q, \quad y \mapsto 0 \text{ in } R, \qquad x \in P, \quad x \mapsto 0 \text{ in } Q_y.$$

Every such x is killed in R, so there is a canonical map to the right side.

Conversely, if $x \in P$ is killed in R, take $y = f(x)$. Then y is killed in R, and x is killed in Q_y. Thus every index on the right occurs on the left. All finite collections of these indices and their comparisons can be accommodated at a common larger pair: replace the x's and y's by sums, and add the image of the chosen larger x to the larger y. Hence any equality in the colimit on the right already holds at a common index on the left. This proves that the canonical map is an isomorphism. □

7 A finite convex-cover calculation

7.1 A nerve lemma in the required form

Lemma 7.1. *Let U be a nonempty convex subset of a finite-dimensional real vector space. Let $D_1, \dots, D_m$ be closed convex polyhedra such that*

$$U = \bigcup_{i=1}^{m} (U \cap D_i).$$

Let N be the nerve whose nonempty simplices are the subsets $I \subset \{1, \dots, m\}$ for which

$$U \cap \bigcap_{i \in I} D_i \neq \varnothing.$$

For every abelian group A, the augmented simplicial cochain complex of N with coefficients A is exact.

Proof. For each simplex I of N, choose a point $u_I \in U \cap \bigcap_{i \in I} D_i$, and let K be the convex hull of these finitely many points. Then $K \subset U$ is a nonempty compact convex polytope. The polytopes $K_i = K \cap D_i$ cover K and have exactly the nerve N.

There is a finite triangulation of K in which every K_i is a subcomplex. To construct one, cut K by the finitely many hyperplanes defining the D_i, take all resulting polyhedral cells and their faces, and barycentrically subdivide this finite polyhedral complex.

For nonempty I, put $K_I = \bigcap_{i \in I} K_i$. Consider the double complex

$$D_{p,q} = \bigoplus_{i_0 < \cdots < i_p} C_q(K_{\{i_0, \dots, i_p\}}; \mathbb{Z}),$$

with simplicial boundary vertically and the alternating intersection maps horizontally. The horizontal complex augmented to $C_q(K; \mathbb{Z})$ is exact: for each simplex of the triangulation, its copies are indexed by the nonempty subsets of the nonempty set of i's whose K_i contain that simplex. This is the augmented chain complex of a simplex.

Every nonempty K_I is convex and hence contractible. Consequently its simplicial homology is $\mathbb{Z}$ in degree zero and zero in higher degrees. Here one may use homotopy invariance and the simplicial–singular comparison [1, Theorems 2.10 and 2.27]. Computing the homology of the finite double complex in its two directions shows that the nerve has the homology of K. The double-complex argument is the standard one recalled in [2, Tag 012X]. Thus the augmented integral chain complex of N is exact.

This is a finite exact complex of finitely generated free abelian groups, including its augmentation to $\mathbb{Z}$. Its cycle groups are free, so its short exact sequences split. Applying $\mathrm{Hom}_{\mathbb{Z}}(-, A)$ therefore preserves exactness and gives the claimed augmented cochain complex. □

7.2 Weak valuative covers

Definition 7.2. A finite family

$$T \longrightarrow T\langle S_i \rangle, \qquad 1 \leq i \leq m,$$

is *weakly valuative* if every homomorphism from T to a valuative monoid factors through at least one member of the family. No infinitesimal lifting condition is imposed in this definition.

Every birhizomic cover is weakly valuative. For a nonempty subset $I \subset \{1, \dots, m\}$, write

$$T_I = T\left\langle \bigcup_{i \in I} S_i \right\rangle.$$

These are the intersections in the opposite site.

Proposition 7.3 (The fs Čech calculation)**.** *Suppose T is fs, $H : B_T \to \mathrm{Ab}$ is arbitrary, and $\{T \to T\langle S_i\rangle\}_{i=1}^m$ is weakly valuative. Then the augmented alternating complex*

$$0 \longrightarrow H(T) \longrightarrow \prod_i (\mathcal{L}_T H)(T_i) \longrightarrow \prod_{i<j} (\mathcal{L}_T H)(T_{\{i,j\}}) \longrightarrow \cdots \longrightarrow (\mathcal{L}_T H)(T_{\{1,\dots,m\}}) \longrightarrow 0 \qquad (5)$$

is exact.

Proof. By Lemma 4.5, we may identify B_T with the finite poset of faces of T. For a face E, let T_E denote T_x for any x with $E_x = E$. If R is a monoid under T, its kernel $E_R = \ker(T \to R)$ is a face, so

$$(\mathcal{L}_T H)(R) = H(T_{E_R}). \qquad (6)$$

Set $L = T^{\mathrm{gp}}$, and consider the polyhedral cones

$$\sigma = \{\ell \in \mathrm{Hom}(L, \mathbb{R}) : \ell(T) \geq 0\}, \qquad \sigma_i = \{\ell \in \sigma : \ell(s) \geq 0 \text{ for all } s \in S_i\}.$$

Weak valuative covering, applied to $\mathbb{R}_{\geq 0}$, implies

$$\sigma = \bigcup_i \sigma_i.$$

For nonempty I, put $\sigma_I = \bigcap_{i \in I} \sigma_i$ and $E_I = \ker(T \to T_I)$. The universal property of rational extensions identifies σ_I with the homomorphisms $T_I \to \mathbb{R}_{\geq 0}$. Lemma 4.6 gives

$$E_I = \{a \in T : \ell(a) = 0 \text{ for every } \ell \in \sigma_I\}. \qquad (7)$$

Indeed, a nonzero element of the fs monoid T_I is detected by a strictly positive real homomorphism.

Fix a face E of T and an abelian group A. Define a functor $J_E(A)$ on the face poset by

$$J_E(A)(T_D) = \begin{cases} A, & D \subset E, \\ 0, & D \not\subset E, \end{cases}$$

with identity transition maps whenever both values are A.

> **Reviewer 2**
>
> worth mentioning/defining as right adjoint to restriction to opens, as that's how it's used

Choose generators $p_1, \dots, p_r$ of T, and put

$$U_E = \{\ell \in \sigma : \ell(p_j) > 0 \text{ for every } p_j \notin E\}.$$

This is nonempty and convex: nonemptiness follows from the strictly positive functional in Lemma 4.6, with the case $T = \{0\}$ interpreted in the evident way.

We claim that

$$E_I \subset E \quad \Longleftrightarrow \quad \sigma_I \cap U_E \neq \varnothing. \qquad (8)$$

For the forward implication, each generator $p_j \notin E$ is outside E_I. By (7), there is an element of σ_I positive on that generator. Summing these finitely many functionals gives an element of $\sigma_I \cap U_E$. If there are no such generators, the zero functional suffices.

Conversely, if $E_I \not\subset E$, express an element of $E_I \setminus E$ in the chosen generators. Since E_I is a face, every generator occurring belongs to E_I; since E is a face, at least one occurring generator lies outside E. Every functional in σ_I vanishes on that generator, so $\sigma_I \cap U_E = \varnothing$.

For $H = J_E(A)$, equations (6) and (8) identify (5) with the augmented cochain complex of the nerve of

$$U_E = \bigcup_i (U_E \cap \sigma_i).$$

It is exact by Lemma 7.1.

It remains to pass from these functors to arbitrary H. The functor J_E is right adjoint to evaluation at T_E:

$$\mathrm{Hom}(H, J_E(A)) = \mathrm{Hom}(H(T_E), A).$$

Thus $J_E(A)$ is injective when A is an injective abelian group. There are enough such injectives: choose embeddings $H(T_E) \hookrightarrow A_E$ into injective abelian groups and use the adjunctions to obtain

$$H \hookrightarrow \prod_E J_E(A_E).$$

The component indexed by $E = D$ proves injectivity at T_D. The existence of injective embeddings of abelian groups is standard; see [2, Tag 01D6]. Because the face poset is finite, every injective functor is a direct summand of a finite product of functors of this form. Consequently (5) is exact for injective H.

Finally, all terms of (5), including its augmentation, are exact functors of H, by (6). Apply them to an injective resolution of H. The resulting double complex has exact augmented rows, while its vertical cohomology is the augmented complex for H. Its horizontal direction is bounded, and, after shifting the augmented degree by one, it is a first-quadrant double complex. Computing total cohomology in the two directions proves exactness for arbitrary H; see again [2, Tag 012X]. □

8 Passage to arbitrary monoids

Lemma 8.1. *Let $T = \bigcup_\lambda T_\lambda$ be a filtered union of fs submonoids. Suppose finite sets $S_1, \dots, S_m \subset T^{\mathrm{gp}}$ define a weakly valuative family over T. After restricting to a cofinal set of indices, all S_i lie in T_λ^{gp}, and*

$$T_\lambda \longrightarrow T_\lambda\langle S_i\rangle, \qquad 1 \le i \le m,$$

is weakly valuative for every remaining λ.

Proof. The finite sets S_i belong to a common T_λ^{gp}, so we first restrict to such indices.

The compact spaces of Lemma 5.3 satisfy

$$K_T \simeq \varprojlim_\lambda K_{T_\lambda}.$$

Indeed, the groups T_λ^{gp} have union T^{gp}, and compatible positive cones on them give exactly a positive cone containing T.

In K_{T_λ}, put

$$D_{i,\lambda} = \{W : S_i \subset W\}, \qquad Z_\lambda = K_{T_\lambda} \setminus \bigcup_i D_{i,\lambda}.$$

These are clopen sets, and the Z_λ form an inverse system of compact spaces. If every Z_λ were nonempty, their inverse limit would be nonempty: every finite collection of compatibility requirements can be satisfied by choosing a point at a common upper index, and compactness then supplies a compatible family. Such a family would define a point of K_T lying in none of the D_i, contrary to weak valuative covering over T.

Thus $Z_\lambda = \varnothing$ at some stage and at every later stage. By Lemma 4.3(i), this is precisely the desired weak valuative covering condition. □

Proposition 8.2 (The general Čech calculation)**.** *Proposition 7.3 holds for an arbitrary monoid T.*

Proof. Choose a filtered union $T = \bigcup_\lambda T_\lambda$ as in Lemma 4.6. By Lemma 8.1, we may suppose that the given finite rational family is defined and weakly valuative at every stage.

For $x \in T_\lambda$, base change gives

$$(T_\lambda)_x \amalg_{T_\lambda} T \simeq T_x.$$

Define $H_\lambda : B_{T_\lambda} \to \mathrm{Ab}$ by

$$H_\lambda((T_\lambda)_x) = H(T_x).$$

For nonempty $I \subset \{1, \dots, m\}$, write

$$T_{I,\lambda} = T_\lambda \left\langle \bigcup_{i \in I} S_i \right\rangle, \qquad T_I = T \left\langle \bigcup_{i \in I} S_i \right\rangle.$$

The universal property of rational extensions gives

> **Reviewer 2**
>
> only defined "finite rational extensions earlier"

$$T_I = \mathrm{colim}_\lambda T_{I,\lambda}.$$

Let $E_{I,\lambda} = \ker(T_\lambda \to T_{I,\lambda})$, and choose $x_{I,\lambda} \in T_\lambda$ generating this face

> **Reviewer 2**
>
> didn't expressly state that T_λ are fs, needed for this assumption, but certainly what was meant

in the sense of Lemma 4.5. Then

$$(\mathcal{L}_{T_\lambda} H_\lambda)(T_{I,\lambda}) = H(T_{x_{I,\lambda}}).$$

There is a natural isomorphism

$$(\mathcal{L}_T H)(T_I) \simeq \mathrm{colim}_\lambda (\mathcal{L}_{T_\lambda} H_\lambda)(T_{I,\lambda}). \tag{9}$$

To verify it, an element $a \in T$ killed in T_I belongs to some T_λ and becomes zero in a later $T_{I,\mu}$, by Lemma 4.1. It therefore belongs to $E_{I,\mu}$, so $a \le n x_{I,\mu}$ in T_μ for some n. Hence the objects $T_{x_{I,\lambda}}$ are cofinal in the indexing category defining $(\mathcal{L}_T H)(T_I)$. The same reasoning gives the compatibility maps and proves (9).

At the augmentation, $H_\lambda(T_\lambda) = H(T)$. There are only finitely many terms in each degree of the alternating Čech complex. Therefore (9) identifies the augmented complex over T with the filtered colimit of the augmented complexes over T_λ. The latter are exact by Proposition 7.3. Filtered colimits of abelian groups are exact, and they commute with finite products, so the complex over T is exact as well; for exactness of filtered colimits, see [2, Tag 00DB]. □

Remark 8.3. Only weak valuative covering was descended in this argument. There is no need to descend the infinitesimal lifting condition, and there is no finite-presentation assumption on the coefficient presheaf.

9 From Čech exactness to acyclicity

We include the sheaf-theoretic argument to make explicit why the preceding calculations imply vanishing of derived, rather than merely Čech, cohomology.

Lemma 9.1. *Let $\mathcal{C}$ be a site admitting a basis of finite covering families by monomorphisms, stable under base change and composition, with the corresponding finite intersections. Suppose an abelian presheaf $\mathcal{E}$ has exact augmented alternating Čech complex for every covering family in this basis. Then $\mathcal{E}$ is a sheaf and*

$$H^q(U, \mathcal{E}) = 0 \qquad \text{for every } U \in \mathcal{C},\ q > 0.$$

> **Reviewer 2**
>
> standard. Nice of the AI to include a proof, but a citation would've sufficed

Proof. Exactness in augmented degrees $-1, 0$ is the sheaf condition for the basis. It implies the sheaf condition for all covers: refine a cover by a basis cover to glue, and use base changes of that basis cover to verify that the glued section restricts to the originally specified sections.

We first recall why an injective abelian sheaf $\mathcal{I}$ has exact augmented alternating Čech complex for a finite cover by monomorphisms. Let $\mathbb{Z}[U]$ denote the free abelian sheaf on the associated representable sheaf of U. The augmented chain complex

$$\cdots \longrightarrow \bigoplus_{i<j} \mathbb{Z}[U_i \times_U U_j] \longrightarrow \bigoplus_i \mathbb{Z}[U_i] \longrightarrow \mathbb{Z}[U] \longrightarrow 0$$

is exact. Locally over a section of U, one of the covering subobjects is available, and inserting its index contracts the augmented complex. This local exactness is exactness after sheafification. Applying the exact contravariant functor $\mathrm{Hom}(-, \mathcal{I})$, and using $\mathrm{Hom}(\mathbb{Z}[U], \mathcal{I}) = \mathcal{I}(U)$, proves the assertion. This is also the usual injective Čech-acyclicity statement; see [2, Tag 03AW].

Embed $\mathcal{E}$ in an injective abelian sheaf $\mathcal{I}$, and let $\mathcal{Q} = \mathcal{I}/\mathcal{E}$. Such embeddings exist on any site [2, Tag 01DP]. We claim that

$$0 \longrightarrow \mathcal{E}(U) \longrightarrow \mathcal{I}(U) \longrightarrow \mathcal{Q}(U) \longrightarrow 0 \tag{10}$$

is exact for every object U.

> **Reviewer 2**
>
> meaning exactness as presheaves

Indeed, a section of $\mathcal{Q}(U)$ lifts locally to $\mathcal{I}$, and we may choose a finite basis cover for these lifts. Local lifting for epimorphisms of abelian sheaves is part of the usual description of sheaf cokernels [2, Tag 03CM]. The differences of the local lifts form a Čech 1-cocycle with values in $\mathcal{E}$. By the assumed exactness, this is a coboundary. Adjusting the lifts by a 0-cochain in $\mathcal{E}$ makes them agree, and the sheaf condition for $\mathcal{I}$ glues them to a global lift. This proves (10).

For every basis cover, (10) yields a short exact sequence of augmented Čech complexes. The complexes for $\mathcal{E}$ and $\mathcal{I}$ are exact, so the complex for $\mathcal{Q}$ is exact. Thus $\mathcal{Q}$ has the same property as $\mathcal{E}$.

The long exact sequence of the right derived functors of evaluation, applied to $0 \to \mathcal{E} \to \mathcal{I} \to \mathcal{Q} \to 0$, gives

$$H^1(U, \mathcal{E}) = 0, \qquad H^q(U, \mathcal{E}) \simeq H^{q-1}(U, \mathcal{Q}) \quad (q \geq 2).$$

Repeating the argument with $\mathcal{Q}$ proves vanishing in every positive degree. □

10 Proof of the theorem

Proof of Theorem 3.1. Let

$$\mathcal{G}_F = \mathcal{L}_P F.$$

Lemma 6.1 proves formula (1) and identifies its restriction to B_P with F.

Consider any object $P \to Q$. By Proposition 5.4, every birhizomic cover of this object has a refinement of the form

$$Q \longrightarrow Q\langle S_i \rangle, \qquad 1 \le i \le m,$$

which is a finite birhizomic cover. These finite rational covers form a basis, and their opposites are monomorphisms by Remark 4.4.

Define $H : B_Q \to \mathrm{Ab}$ by

$$H(Q_y) = \mathcal{G}_F(P \to Q \to Q_y).$$

Lemma 6.2 identifies $\mathcal{G}_F$, on monoids under Q, with $\mathcal{L}_Q H$. The displayed covering family is weakly valuative. Consequently, Proposition 8.2 proves exactness of its augmented alternating Čech complex with coefficients in $\mathcal{G}_F$.

This applies to every object Q and every basis cover. Lemma 9.1 therefore gives both

$$\mathcal{G}_F \in \mathrm{Sh}(A_P^{\mathrm{op}}, \mathrm{Ab}) \quad \text{and} \quad H^q(Q, \mathcal{G}_F) = 0 \quad (q > 0).$$

Since $\mathcal{G}_F$ is the presheaf left Kan extension and is already a sheaf, it is canonically $\pi^{-1}F$. This proves (i) and (ii).

Finally, the right adjoint π_* is restriction to B_P. Indeed, restriction is right adjoint to presheaf left Kan extension, and sheafification is left adjoint to inclusion of sheaves into presheaves. The topology on B_P^{op} is trivial, so no additional sheafification is involved on that side.

Take an injective resolution $\mathcal{G}_F \to \mathcal{I}^\bullet$ in abelian sheaves on A_P^{op}. Because cohomology in the presheaf category on B_P is computed objectwise, for every $x \in P$ we have

$$\big(\mathrm{R}^q \pi_* \mathcal{G}_F\big)(P_x) = H^q\big(\mathcal{I}^\bullet(P_x)\big) = H^q(P_x, \mathcal{G}_F).$$

These groups vanish for $q > 0$. In degree zero, the canonical left Kan extension unit gives

$$(\pi_* \mathcal{G}_F)(P_x) = \mathcal{G}_F(P_x) = F(P_x).$$

Thus the derived unit induces the identity isomorphism in degree zero and has zero higher cohomology on its target. It is therefore a quasi-isomorphism:

$$\boxed{F \xrightarrow{\sim} \mathrm{R}\pi_* \pi^{-1} F}.$$

□

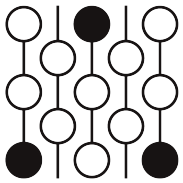
References

[1] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Theorems 2.10 and 2.27.

[2] The Stacks Project Authors, *The Stacks Project*. Tags `00DB` (exactness of filtered colimits), `012X` (spectral sequences of double complexes), `01D6` (injective abelian groups), `01DP` (enough injectives for abelian sheaves on a site), `03CM` (abelian sheaves and local exactness), and `03AW` (injective Čech acyclicity).

Review of Problem 1 – Astra

First Proof Third Batch
Reviewer 3
October 8–9, 2026



1 Review

Global opinion: it has a good strategy, and solves the problem fully. It uses standard tools, no novelty in the arguments compared to the human solution. It sets up more structure than the human solution, which I personally enjoy, and I think makes it easier to follow in this aspect. (Shared with Lynx)
It fails almost everywhere to explain the structure of its proofs. Most of them are verifiable line-by-line, but only understandable to someone who already understands what's going on.
Attribution is fine; at some point is it a bit too imprecise (citing a Sections with lots of results instead of a specific result), which feels quite sloppy.

Correctness. Essentially flawless.

One small mistake: it defines "weakly valuative" in one context, but uses it in a slightly larger context. The correct definition in the larger context is literally the same, so it is still essentially flawless in my opinion.
Score: 1

Exposition. Line-by-line ok to verify, but missing signposting/overview of proofs. Making most lines very difficult to understand the meaning/significance of.

E.g., the proof of 7.3 contains lots of steps, and I agree with the proof after reading all of them, but I'd like a high-level overview such as "we will prove it first for these special values of H s and then show every value of H can be approximated by the special values."
Score: 0

Attributions. It is very self-contained. The citations it does make check out, for the rest it's mostly doing rather elementary homological/monoid algebra.

The Stacks Project citations could be more specific. The Stacks Project specifically allows for citing very precisely in a short tag each and every chapter, section, theorem or lemma, and it sometimes cites an entire section when it needs a specific theorem/lemma in it. You could call this correct, but if I'd be the referee, I'd ask them to correct it.

It could cite Ogus for some of it, but I'd argue with it not needing to, especially in the context of answering a research question that treats these concepts as elementary.
Score: 1

2 Submission with inline comments

Abstract

The answer to the question is affirmative, without a finite-generation hypothesis on the monoid. We prove a stronger statement: the presheaf left Kan extension of the given presheaf is already a birhizomic sheaf and is acyclic on every object of the birhizomic site. The proof uses finite refinements of birhizomic covers, a finite convex-cover calculation, and passage to filtered unions of finitely generated saturated monoids.

3 Statement and conventions

We use the definition of monoid in the question. In intermediate constructions, submonoids of groups are allowed to have units; we explicitly sharpen them before regarding them as objects of the category in the question. All group-valued expressions involving a monoid homomorphism refer to its extension to groupifications.

Put

$$P_x = P[-x]^\sharp \qquad (x \in P).$$

A presheaf on B_P^{op} will be regarded as a covariant functor $B_P \to \mathrm{Ab}$. Likewise, presheaves on A_P^{op} are covariant functors on A_P.

For a sheaf $\mathcal{E}$ on A_P^{op}, write $H^q(Q, \mathcal{E})$ for the right derived functors of evaluation at the object $P \to Q$. We make the usual universe choices for sites and sheaf categories.

Theorem 3.1. *Let P be any monoid, and let $F : B_P \to \mathrm{Ab}$ be any functor. For $f : P \to Q$, define*

$$\mathcal{G}_F(Q) = \operatorname{colim}_{\substack{x \in P \\ f(x)=0}} F(P_x), \tag{1}$$

where the indexing category consists of the corresponding objects P_x and their morphisms over P. Then:

(i) *$\mathcal{G}_F$ is the presheaf left Kan extension of F, and it is already a sheaf for the birhizomic topology;*

(ii) *for every $P \to Q$,*

$$H^q(Q, \mathcal{G}_F) = 0 \qquad (q > 0);$$

(iii) *the canonical unit is a quasi-isomorphism:*

$$F \xrightarrow{\sim} \mathrm{R}\pi_* \pi^{-1} F.$$

The indexing category in (1) is filtered, as will be proved below. In particular, the formula does not require its indexing face to be finitely generated.

4 Elementary constructions with monoids

4.1 Filtered colimits and finite rational extensions

> **Reviewer 3**
>
> Exp: “commutative monoids” is unclear in what follows.

Lemma 4.1. *The groupification of a monoid is torsion-free. Filtered colimits of monoids are computed on the underlying commutative monoids, and hence on the underlying sets.*

Proof. If $ng = 0$ in P^{gp}, with $n > 0$, saturation gives $g, -g \in P$, and sharpness gives $g = 0$.

Let $P = \operatorname{colim}_\lambda P_\lambda$ be a filtered colimit of the underlying commutative monoids. Integrality and sharpness hold in P: any equality witnessing a failure of either property holds at some stage, where that property is valid.

To check saturation, write an element of P^{gp} as $a-b$, with $a, b \in P$, and suppose $n(a-b) = c \in P$. By integrality, this is the equality $na = nb + c$ in P. Representatives for a, b, c, together with this equality, exist at a common stage. Saturation at that stage shows that $a - b$ belongs to its positive monoid, and therefore belongs to P. □

Definition 4.2. For a finite subset $S \subset P^{\mathrm{gp}}$, define

$$C_P(S) = \left\{ g \in P^{\mathrm{gp}} : ng \in P + \sum_{s \in S} \mathbb{N}s \text{ for some } n > 0 \right\},$$

and put

$$P\langle S\rangle = C_P(S)/\big(C_P(S) \cap (-C_P(S))\big).$$

We call $P \to P\langle S\rangle$ a *finite rational extension*. This terminology is used only for the construction just specified.

Lemma 4.3. *The construction in Definition 4.2 has the following properties.*

(i) *For any $h : P \to M$, there is a morphism $P\langle S\rangle \to M$ extending h if and only if $h(s) \in M$ for all $s \in S$. Such a morphism is unique.*

(ii) *The morphism $P \to P\langle S\rangle$ is of finite presentation, and its groupification is surjective.*

(iii) *Finite rational extensions are preserved by arbitrary pushout and by composition. Moreover,*

$$P\langle S\rangle \amalg_P P\langle T\rangle \simeq P\langle S \cup T\rangle.$$

(iv) *$P\langle\{-x\}\rangle \simeq P_x$ for $x \in P$.*

Proof. The submonoid $C = C_P(S)$ is saturated in P^{gp}, and its units form the subgroup $U = C \cap (-C)$. The quotient C/U is saturated: if $n(g + U)$ belongs to C/U, then $ng \in C$, since adding or subtracting an element of U does not change membership in C. Thus $g \in C$. The quotient is integral and sharp by construction.

If $h(s) \in M$ for all $s \in S$, saturation of M implies that $h(C) \subset M$, and sharpness implies that $h(U) = 0$. This gives the asserted extension. Its uniqueness follows because $P^{\mathrm{gp}} \to (P\langle S\rangle)^{\mathrm{gp}}$ is surjective.

Write $s = a_s - b_s$, with $a_s, b_s \in P$. The condition $h(s) \in M$ is equivalent to the existence of an element $z_s \in M$ such that

$$h(b_s) + z_s = h(a_s).$$

There are finitely many such conditions, and each z_s, when it exists, is unique by integrality. Lemma 4.1 therefore shows that $\mathrm{Hom}_P(P\langle S\rangle, -)$ commutes with filtered colimits. This proves finite presentation.

The pushout assertion follows immediately from (i). More generally, under $h : P \to Q$, the pushout is $Q\langle h(S)\rangle$. For composition, lift the finitely many group elements used in a rational extension of $P\langle S\rangle$ to P^{gp}; adjoining these lifts together with S gives the composite.

Finally, $h(-x) \in M$, with $h(x) \in M$, is equivalent to $h(x) = 0$, since M is sharp. This is precisely the universal property of P_x. □

Remark 4.4. Every finite rational extension is an epimorphism in the category of monoids, by the uniqueness assertion in Lemma 4.3(i). Consequently, its opposite is a monomorphism of the site. This observation will allow us to use finite alternating Čech complexes.

4.2 Faces and the category B_P

A *face* of P is a submonoid $E \subset P$ such that $a + b \in E$ implies $a, b \in E$. The inverse image of 0 under a homomorphism of sharp monoids is a face.

Lemma 4.5. *For $x \in P$, set*

$$E_x = \{a \in P : a \leq nx \text{ for some } n \in \mathbb{N}\}.$$

Then $E_x = \ker(P \to P_x)$, and

$$\mathrm{Hom}_P(P_x, P_y) \neq \varnothing \quad \Longleftrightarrow \quad E_x \subset E_y.$$

When this set of morphisms is nonempty, it is a singleton.

If P is finitely generated, every face is of the form E_x, there are only finitely many faces, and B_P, up to equivalence, is their inclusion poset.

Proof. The localization $P + \mathbb{N}(-x) \subset P^{\mathrm{gp}}$ is already saturated. Indeed, if $ng = a - kx$, with $a \in P$ and $k \geq 0$, then

$$n(g + kx) = a + (n-1)kx \in P,$$

so $g + kx \in P$. An element $a \in P$ becomes a unit in this localization precisely when $a \leq nx$ for some n. This proves the kernel formula. The assertion about morphisms follows from the universal property of P_x.

If $p_1, \ldots, p_r$ generate P, a face E is generated by those p_j which belong to E: every summand in an expression for an element of a face belongs to that face. Put

$$x_E = \sum_{p_j \in E} p_j.$$

Every element of E is bounded above by a multiple of x_E, and the face property gives $E_{x_E} \subset E$. Hence $E = E_{x_E}$. The same description proves finiteness of the set of faces. □

4.3 Finite-dimensional approximation

We call a monoid *fs* if it is finitely generated, in addition to the standing assumptions.

Lemma 4.6. *The following assertions hold.*

(i) *Every monoid is a filtered union of fs submonoids.*

(ii) *If P is fs and $S \subset P^{\mathrm{gp}}$ is finite, then $P\langle S\rangle$ is fs.*

(iii) *If P is fs, there is a homomorphism $\ell : P \to \mathbb{R}_{\geq 0}$ which is strictly positive on $P \setminus \{0\}$.*

Proof. We first recall the relevant elementary lattice calculation. Let L be a finitely generated free abelian group, and let $v_1, \ldots, v_r \in L$. If

$$C = \sum_j \mathbb{R}_{\geq 0} v_j,$$

then the saturation in L of $\sum_j \mathbb{N} v_j$ is $C \cap L$. Indeed, a nonnegative real solution to a linear system with rational coefficients and rational right-hand side can be replaced by a nonnegative rational solution. One way to see this is to choose a solution with minimal support: dependence among its supporting columns permits movement until a coordinate becomes zero; with independent supporting columns, the solution is rational by linear algebra. Clearing denominators proves the assertion.

> Reviewer 3
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> Exp: a bit overkill to spend a paragraph on a simple linear algebra fact.

Furthermore, $C \cap L$ is finitely generated. For $g = \sum_j t_j v_j \in C \cap L$, subtract $\sum_j \lfloor t_j \rfloor v_j$. The remainder is a lattice point in the bounded set

$$\left\{ \sum_j u_j v_j : 0 \le u_j \le 1 \right\}.$$

There are only finitely many such remainders, and they, together with the v_j, generate $C \cap L$.

For a finite subset of an arbitrary monoid P, take the saturation of the submonoid it generates in the subgroup it generates in P^{gp}. That subgroup is a finitely generated torsion-free abelian group, hence is free. The saturation lies in P, is finitely generated by the preceding calculation, and is sharp. These submonoids are directed under inclusion and exhaust P. This proves (i). Applying the same calculation to generators of P together with S, and then taking the quotient by units, proves (ii).

For (iii), assume $P \neq \{0\}$, and choose nonzero generators $p_1, \dots, p_r$. The origin does not belong to their convex hull. Otherwise the rational-solution argument above would give a nontrivial relation

$$\sum_j n_j p_j = 0, \qquad n_j \in \mathbb{N},$$

contrary to sharpness. Choose an inner product on $P^{\mathrm{gp}} \otimes \mathbb{R}$, and let z be the point in this compact convex hull nearest the origin. Then $z \neq 0$, and minimality gives

$$\langle z, p_j \rangle \ge \|z\|^2 > 0$$

for every j. The functional $u \mapsto \langle z, u \rangle$ has the required property. For $P = \{0\}$, the assertion is immediate. □

5 Infinitesimal envelopes and finite covering refinements

5.1 An explicit infinitesimal envelope

Lemma 5.1. *Let $v : P \to V$ be a homomorphism to a valuative monoid. There is a factorization*

$$P \longrightarrow M_v \xrightarrow{p_v} V$$

such that p_v is infinitesimal and which is initial among factorizations of v whose second morphism is infinitesimal.

More explicitly, with $G = P^{\mathrm{gp}}$, put

$$C_v = \left\{ a \in G : \begin{array}{l} \text{there exist } y \in P,\ c \in \mathbb{N} \text{ such that} \\ a \prec y, \quad v(a) \ge 0, \quad v(y) \le c\, v(a) \end{array} \right\}. \tag{2}$$

Then

$$M_v = C_v/(C_v \cap (-C_v)).$$

Moreover,

$$M_v = \operatorname{colim}_{\substack{S \subset C_v \\ S \text{ finite}}} P\langle S \rangle \tag{3}$$

as monoids under P.

Proof. For a nonnegative element y, the condition $a \prec y$ can equivalently be written

$$-ky \leq a \leq ky$$

for some $k \in \mathbb{N}$. Also, for nonnegative b, d, the condition $b \prec d$ is equivalent to $b \leq cd$ for some $c \in \mathbb{N}$.

The set C_v contains P. It is closed under addition: if a, a' have witnesses y, y', then $a+a' \prec y+y'$, and, after increasing the two integer bounds to a common bound, $v(y+y')$ is bounded by a multiple of $v(a+a')$.

It is saturated. Suppose $na \in C_v$, with witness y. Choose k such that $ky \pm na \in P$. Since

$$n(ky \pm a) = (ky \pm na) + (n-1)ky \in P,$$

saturation of P gives $ky \pm a \in P$. Saturation of V gives $v(a) \geq 0$, and the inequality bounding $v(y)$ by a multiple of $nv(a)$ is also a bound by a multiple of $v(a)$.

Let $E = v^{-1}(0)$. We claim that

$$C_v \cap (-C_v) = E^{\mathrm{gp}}. \tag{4}$$

If $a, -a \in C_v$, then $v(a) = 0$. A witness y for a therefore satisfies $v(y) = 0$. For sufficiently large k, $a + ky \in P$, and it too has image zero. Thus $a = (a+ky) - ky \in E^{\mathrm{gp}}$. Conversely, a difference of two elements of E is bounded in absolute value by their sum and satisfies (2) in both signs. This proves (4). Sharpening C_v gives a monoid M_v and a well-defined map $p_v : M_v \to V$.

We next verify that p_v is infinitesimal. Let $a \in M_v^{\mathrm{gp}}$ and $b \in M_v$ satisfy

$$p_v(a) \geq 0, \qquad a \prec b, \qquad p_v(b) \prec p_v(a).$$

Choose representatives $x \in G$ and $z \in C_v$ of a, b. A lower bound in $a \prec b$ gives an integer m such that

$$d = x - mz \in C_v.$$

Here an equality modulo the unit subgroup can be lifted in this way because adding or subtracting a unit preserves C_v.

Choose witnesses $y_d, y_z \in P$ for d, z. The inequality $p_v(b) \prec p_v(a)$ gives $v(z) \leq e\, v(x)$ for some e. Since $v(d) = v(x) - mv(z) \geq 0$, both $v(d)$ and $v(z)$ are bounded by multiples of $v(x)$. Hence $v(y_d + y_z)$ is bounded by a multiple of $v(x)$. On the other hand,

$$x = d + mz \prec y_d + y_z$$

in the original ordered group P^{gp}. Thus $x \in C_v$, so $a \in M_v$.

Now consider a factorization

$$P \xrightarrow{h} H \xrightarrow{p} V$$

with p infinitesimal. For $a \in C_v$, a witness y in (2) gives

$$h(a) \prec h(y), \qquad p(h(a)) = v(a) \geq 0, \qquad p(h(y)) \prec p(h(a)).$$

Infinitesimality implies $h(a) \in H$. Thus h^{gp} maps C_v into H, kills its units, and induces $M_v \to H$. This map is unique because $P^{\mathrm{gp}} \to M_v^{\mathrm{gp}}$ is surjective. This proves initiality.

Finally, a homomorphism from the right side of (3) to a monoid H is exactly a homomorphism $P \to H$ carrying every element of C_v into H. By the universal property of sharpening, this is exactly a homomorphism $M_v \to H$. This proves the colimit assertion. □

Corollary 5.2. *If $S \subset C_v$ is finite, then*

$$P \longrightarrow P\langle S \rangle$$

is birhizomic at its induced map to V.

> **Reviewer 3**
>
> *Exp: what is V? [Same as in previous Lemma]*

Proof. Finite presentation follows from Lemma 4.3. For any infinitesimal factorization of v, the proof of Lemma 5.1 shows that every element of S has nonnegative image in the intermediate monoid. Thus the required lift exists by Lemma 4.3(i). It is unique by that lemma, and its composite to V is the specified map, again by uniqueness. □

5.2 Compactness of valuative tests

Lemma 5.3. *Let $G = P^{\mathrm{gp}}$. The set*

$$K_P = \{W \subset G : P \subset W,\ W + W \subset W,\ W \cup (-W) = G\}$$

is a compact Hausdorff space with the topology induced from $\{0,1\}^G$.

Every $W \in K_P$ defines a valuative monoid

$$V_W = W/(W \cap (-W))$$

and a homomorphism $v_W : P \to V_W$. Conversely, every homomorphism from P to a valuative monoid defines a point of K_P. A finite collection of inequalities between group elements defines a clopen subset of K_P.

Proof. All defining conditions on W are closed conditions involving finitely many coordinates at a time. Thus K_P is closed in the compact Hausdorff product $\{0,1\}^G$.

For completeness, compactness of this product follows directly from the ultrafilter argument. Extend a family of closed subsets with the finite intersection property to an ultrafilter, using Zorn's lemma. At each coordinate the ultrafilter chooses exactly one of the two values. Every basic neighborhood of the resulting point belongs to the ultrafilter, so that point belongs to every closed set in the original family.

The submonoid W is saturated. Indeed, if $ng \in W$ and $g \notin W$, then $-g \in W$, whence

$$g = ng + (n-1)(-g) \in W,$$

a contradiction. Quotienting by its units therefore produces a valuative monoid. Conversely, for $v : P \to V$, take

$$W = \{g \in G : v(g) \geq 0\}.$$

An inequality $v(a) \geq v(b)$ is the coordinate condition $a - b \in W$, proving the last assertion. □

Proposition 5.4 (Finite rational refinements)**.** *Every birhizomic cover of a monoid admits a refinement which is a finite birhizomic cover by finite rational extensions.*

These finite covers form a basis of covering families, stable under base change and composition.

Proof. Let $\{P \to Q_i\}_{i\in I}$ be a birhizomic cover. Fix $W \in K_P$. Apply the covering condition to $v_W : P \to V_W$, and choose

$$P \xrightarrow{f_i} Q_i \xrightarrow{g_i} V_W$$

with f_i birhizomic at g_i.

> Reviewer 3
>
> Exp: nested subscripts

The infinitesimal map $M_{v_W} \to V_W$ gives a lift

$$Q_i \longrightarrow M_{v_W}.$$

Since f_i is of finite presentation, (3) implies that this lift factors through $P\langle S_W\rangle$ for some finite $S_W \subset C_{v_W}$. We obtain a refinement morphism

$$Q_i \longrightarrow P\langle S_W\rangle.$$

For each $s \in S_W$, choose a witness $y_s \in P$, an integer $c_s \geq 0$, and bounds $s \prec y_s$, as in (2). Let

$$O_W = \left\{W' \in K_P : s \in W',\ c_s s - y_s \in W' \text{ for all } s \in S_W\right\}.$$

This is a clopen neighborhood of W. For any valuative homomorphism $v : P \to V$ whose positive cone belongs to O_W, the same witnesses show $S_W \subset C_v$. Corollary 5.2 then shows that $P \to P\langle S_W\rangle$ is birhizomic at its induced map to V.

Compactness gives finitely many such neighborhoods covering K_P. The corresponding rational extensions form a birhizomic cover, by the preceding observation for arbitrary valuative targets, and refine the original cover.

Stability under base change and composition follows from Lemma 4.3 and the topology axioms assumed in the question. □

6 The left Kan extension and its transitivity

Lemma 6.1. *For $F : B_P \to \mathrm{Ab}$, its presheaf left Kan extension to A_P is given by* (1)*. Its indexing categories are filtered, and its restriction to B_P is canonically F.*

Proof. By the universal property of P_x, a morphism $P_x \to Q$ over P exists exactly when $f(x) = 0$, and then it is unique. This gives the usual comma-category formula for left Kan extension.

If x, y are killed by f, then $x + y$ is killed by f, and there are morphisms

$$P_x \longrightarrow P_{x+y}, \qquad P_y \longrightarrow P_{x+y}.$$

There is also the object $P_0 = P$, and all morphisms in the indexing category are unique when they exist. Thus the category is filtered.

For $Q = P_x$, the comma category has the terminal object $P_x \xrightarrow{\mathrm{id}} P_x$, proving the restriction assertion. □

We use the notation $\mathcal{L}_P F$ for this presheaf left Kan extension.

Lemma 6.2 (Transitivity)**.** *Fix $f : P \to Q$, and define a presheaf H on B_Q by*

$$H(Q_y) = (\mathcal{L}_P F)(P \to Q \to Q_y).$$

For every $Q \to R$, there is a canonical isomorphism

$$(\mathcal{L}_Q H)(R) \simeq (\mathcal{L}_P F)(R).$$

Proof. Expanding the left side gives a filtered colimit over pairs (y, x) such that

$$y \in Q, \quad y \mapsto 0 \text{ in } R, \qquad x \in P, \quad x \mapsto 0 \text{ in } Q_y.$$

Every such x is killed in R, so there is a canonical map to the right side.

Conversely, if $x \in P$ is killed in R, take $y = f(x)$. Then y is killed in R, and x is killed in Q_y. Thus every index on the right occurs on the left. All finite collections of these indices and their comparisons can be accommodated at a common larger pair: replace the x's and y's by sums, and add the image of the chosen larger x to the larger y. Hence any equality in the colimit on the right already holds at a common index on the left. This proves that the canonical map is an isomorphism. □

7 A finite convex-cover calculation

7.1 A nerve lemma in the required form

Lemma 7.1. *Let U be a nonempty convex subset of a finite-dimensional real vector space. Let $D_1, \dots, D_m$ be closed convex polyhedra such that*

$$U = \bigcup_{i=1}^{m} (U \cap D_i).$$

Let N be the nerve whose nonempty simplices are the subsets $I \subset \{1, \dots, m\}$ for which

$$U \cap \bigcap_{i \in I} D_i \neq \varnothing.$$

For every abelian group A, the augmented simplicial cochain complex of N with coefficients A is exact.

Proof. For each simplex I of N, choose a point $u_I \in U \cap \bigcap_{i \in I} D_i$, and let K be the convex hull of these finitely many points. Then $K \subset U$ is a nonempty compact convex polytope. The polytopes $K_i = K \cap D_i$ cover K and have exactly the nerve N.

There is a finite triangulation of K in which every K_i is a subcomplex. To construct one, cut K by the finitely many hyperplanes defining the D_i, take all resulting polyhedral cells and their faces, and barycentrically subdivide this finite polyhedral complex.

For nonempty I, put $K_I = \bigcap_{i \in I} K_i$. Consider the double complex

$$D_{p,q} = \bigoplus_{i_0 < \cdots < i_p} C_q(K_{\{i_0, \dots, i_p\}}; \mathbb{Z}),$$

with simplicial boundary vertically and the alternating intersection maps horizontally. The horizontal complex augmented to $C_q(K; \mathbb{Z})$ is exact: for each simplex of the triangulation, its copies are

indexed by the nonempty subsets of the nonempty set of i's whose K_i contain that simplex. This is the augmented chain complex of a simplex.

Every nonempty K_I is convex and hence contractible. Consequently its simplicial homology is $\mathbb{Z}$ in degree zero and zero in higher degrees. Here one may use homotopy invariance and the simplicial–singular comparison [1, Theorems 2.10 and 2.27]. Computing the homology of the finite double complex in its two directions shows that the nerve has the homology of K. The double-complex argument is the standard one recalled in [2, Tag 012X]. Thus the augmented integral chain complex of N is exact.

This is a finite exact complex of finitely generated free abelian groups, including its augmentation to $\mathbb{Z}$. Its cycle groups are free, so its short exact sequences split. Applying $\mathrm{Hom}_{\mathbb{Z}}(-, A)$ therefore preserves exactness and gives the claimed augmented cochain complex. □

7.2 Weak valuative covers

Definition 7.2. A finite family

$$T \longrightarrow T\langle S_i\rangle, \qquad 1 \le i \le m,$$

is *weakly valuative* if every homomorphism from T to a valuative monoid factors through at least one member of the family. No infinitesimal lifting condition is imposed in this definition.

> **Reviewer 3**
>
> Exp: the next statement, or the definition above, is wrong. Birhizomic covers need not consist of these finite rational extensions, and hence the definition above apply does not apply to the next statement.

Every birhizomic cover is weakly valuative. For a nonempty subset $I \subset \{1, \dots, m\}$, write

$$T_I = T\left\langle \bigcup_{i\in I} S_i \right\rangle.$$

These are the intersections in the opposite site.

Proposition 7.3 (The fs Čech calculation)**.** *Suppose T is fs, $H : B_T \to \mathrm{Ab}$ is arbitrary, and $\{T \to T\langle S_i\rangle\}_{i=1}^m$ is weakly valuative. Then the augmented alternating complex*

$$0 \longrightarrow H(T) \longrightarrow \prod_i (\mathcal{L}_T H)(T_i) \longrightarrow \prod_{i<j} (\mathcal{L}_T H)(T_{\{i,j\}}) \longrightarrow \cdots \longrightarrow (\mathcal{L}_T H)(T_{\{1,\dots,m\}}) \longrightarrow 0 \tag{5}$$

is exact.

Proof. By Lemma 4.5, we may identify B_T with the finite poset of faces of T. For a face E, let T_E denote T_x for any x with $E_x = E$. If R is a monoid under T, its kernel $E_R = \ker(T \to R)$ is a face, so

$$(\mathcal{L}_T H)(R) = H(T_{E_R}). \tag{6}$$

Set $L = T^{\mathrm{gp}}$, and consider the polyhedral cones

$$\sigma = \{\ell \in \mathrm{Hom}(L, \mathbb{R}) : \ell(T) \ge 0\}, \qquad \sigma_i = \{\ell \in \sigma : \ell(s) \ge 0 \text{ for all } s \in S_i\}.$$

Weak valuative covering, applied to $\mathbb{R}_{\ge 0}$, implies

$$\sigma = \bigcup_i \sigma_i.$$

For nonempty I, put $\sigma_I = \bigcap_{i\in I} \sigma_i$ and $E_I = \ker(T \to T_I)$. The universal property of rational extensions identifies σ_I with the homomorphisms $T_I \to \mathbb{R}_{\geq 0}$. Lemma 4.6 gives

$$E_I = \{a \in T : \ell(a) = 0 \text{ for every } \ell \in \sigma_I\}. \tag{7}$$

Indeed, a nonzero element of the fs monoid T_I is detected by a strictly positive real homomorphism.

Fix a face E of T and an abelian group A. Define a functor $J_E(A)$ on the face poset by

$$J_E(A)(T_D) = \begin{cases} A, & D \subset E, \\ 0, & D \not\subset E, \end{cases}$$

with identity transition maps whenever both values are A.

Choose generators $p_1, \ldots, p_r$ of T, and put

$$U_E = \{\ell \in \sigma : \ell(p_j) > 0 \text{ for every } p_j \notin E\}.$$

This is nonempty and convex: nonemptiness follows from the strictly positive functional in Lemma 4.6, with the case $T = \{0\}$ interpreted in the evident way.

We claim that

$$E_I \subset E \quad \Longleftrightarrow \quad \sigma_I \cap U_E \neq \varnothing. \tag{8}$$

For the forward implication, each generator $p_j \notin E$ is outside E_I. By (7), there is an element of σ_I positive on that generator. Summing these finitely many functionals gives an element of $\sigma_I \cap U_E$. If there are no such generators, the zero functional suffices.

Conversely, if $E_I \not\subset E$, express an element of $E_I \setminus E$ in the chosen generators. Since E_I is a face, every generator occurring belongs to E_I; since E is a face, at least one occurring generator lies outside E. Every functional in σ_I vanishes on that generator, so $\sigma_I \cap U_E = \varnothing$.

For $H = J_E(A)$, equations (6) and (8) identify (5) with the augmented cochain complex of the nerve of

$$U_E = \bigcup_i (U_E \cap \sigma_i).$$

It is exact by Lemma 7.1.

It remains to pass from these functors to arbitrary H. The functor J_E is right adjoint to evaluation at T_E:

$$\operatorname{Hom}(H, J_E(A)) = \operatorname{Hom}(H(T_E), A).$$

Thus $J_E(A)$ is injective when A is an injective abelian group. There are enough such injectives: choose embeddings $H(T_E) \hookrightarrow A_E$ into injective abelian groups and use the adjunctions to obtain

$$H \hookrightarrow \prod_E J_E(A_E).$$

The component indexed by $E = D$ proves injectivity at T_D. The existence of injective embeddings of abelian groups is standard; see [2, Tag 01D6]. Because the face poset is finite, every injective functor is a direct summand of a finite product of functors of this form.

Consequently (5) is exact for injective H.

Finally, all terms of (5), including its augmentation, are exact functors of H, by (6). Apply them to an injective resolution of H. The resulting double complex has exact augmented rows, while its vertical cohomology is the augmented complex for H. Its horizontal direction is bounded, and, after shifting the augmented degree by one, it is a first-quadrant double complex. Computing total cohomology in the two directions proves exactness for arbitrary H; see again [2, Tag 012X]. □

8 Passage to arbitrary monoids

Lemma 8.1. *Let $T = \bigcup_\lambda T_\lambda$ be a filtered union of fs submonoids. Suppose finite sets $S_1, \dots, S_m \subset T^{\mathrm{gp}}$ define a weakly valuative family over T. After restricting to a cofinal set of indices, all S_i lie in T_λ^{gp}, and*

$$T_\lambda \longrightarrow T_\lambda\langle S_i\rangle, \qquad 1 \le i \le m,$$

is weakly valuative for every remaining λ.

Proof. The finite sets S_i belong to a common T_λ^{gp}, so we first restrict to such indices.

The compact spaces of Lemma 5.3 satisfy

$$K_T \simeq \varprojlim_\lambda K_{T_\lambda}.$$

Indeed, the groups T_λ^{gp} have union T^{gp}, and compatible positive cones on them give exactly a positive cone containing T.

In K_{T_λ}, put

$$D_{i,\lambda} = \{W : S_i \subset W\}, \qquad Z_\lambda = K_{T_\lambda} \setminus \bigcup_i D_{i,\lambda}.$$

These are clopen sets, and the Z_λ form an inverse system of compact spaces. If every Z_λ were nonempty, their inverse limit would be nonempty: every finite collection of compatibility requirements can be satisfied by choosing a point at a common upper index, and compactness then supplies a compatible family. Such a family would define a point of K_T lying in none of the D_i, contrary to weak valuative covering over T.

Thus $Z_\lambda = \varnothing$ at some stage and at every later stage. By Lemma 4.3(i), this is precisely the desired weak valuative covering condition. □

Proposition 8.2 (The general Čech calculation)**.** *Proposition 7.3 holds for an arbitrary monoid T.*

Proof. Choose a filtered union $T = \bigcup_\lambda T_\lambda$ as in Lemma 4.6. By Lemma 8.1, we may suppose that the given finite rational family is defined and weakly valuative at every stage.

For $x \in T_\lambda$, base change gives

$$(T_\lambda)_x \amalg_{T_\lambda} T \simeq T_x.$$

Define $H_\lambda : B_{T_\lambda} \to \mathrm{Ab}$ by

$$H_\lambda((T_\lambda)_x) = H(T_x).$$

For nonempty $I \subset \{1, \dots, m\}$, write

$$T_{I,\lambda} = T_\lambda \left\langle \bigcup_{i\in I} S_i \right\rangle, \qquad T_I = T \left\langle \bigcup_{i\in I} S_i \right\rangle.$$

The universal property of rational extensions gives

$$T_I = \mathrm{colim}_\lambda\, T_{I,\lambda}.$$

Let $E_{I,\lambda} = \ker(T_\lambda \to T_{I,\lambda})$, and choose $x_{I,\lambda} \in T_\lambda$ generating this face in the sense of Lemma 4.5. Then

$$(\mathcal{L}_{T_\lambda} H_\lambda)(T_{I,\lambda}) = H(T_{x_{I,\lambda}}).$$

There is a natural isomorphism

$$(\mathcal{L}_T H)(T_I) \simeq \operatorname{colim}_\lambda (\mathcal{L}_{T_\lambda} H_\lambda)(T_{I,\lambda}). \tag{9}$$

To verify it, an element $a \in T$ killed in T_I belongs to some T_λ and becomes zero in a later $T_{I,\mu}$, by Lemma 4.1. It therefore belongs to $E_{I,\mu}$, so $a \leq n x_{I,\mu}$ in T_μ for some n. Hence the objects $T_{x_{I,\lambda}}$ are cofinal in the indexing category defining $(\mathcal{L}_T H)(T_I)$. The same reasoning gives the compatibility maps and proves (9).

At the augmentation, $H_\lambda(T_\lambda) = H(T)$. There are only finitely many terms in each degree of the alternating Čech complex. Therefore (9) identifies the augmented complex over T with the filtered colimit of the augmented complexes over T_λ. The latter are exact by Proposition 7.3. Filtered colimits of abelian groups are exact, and they commute with finite products, so the complex over T is exact as well; for exactness of filtered colimits, see [2, Tag 00DB]. □

Remark 8.3. Only weak valuative covering was descended in this argument. There is no need to descend the infinitesimal lifting condition, and there is no finite-presentation assumption on the coefficient presheaf.

9 From Čech exactness to acyclicity

We include the sheaf-theoretic argument to make explicit why the preceding calculations imply vanishing of derived, rather than merely Čech, cohomology.

Lemma 9.1. *Let $\mathcal{C}$ be a site admitting a basis of finite covering families by monomorphisms, stable under base change and composition, with the corresponding finite intersections. Suppose an abelian presheaf $\mathcal{E}$ has exact augmented alternating Čech complex for every covering family in this basis. Then $\mathcal{E}$ is a sheaf and*

$$H^q(U, \mathcal{E}) = 0 \qquad \text{for every } U \in \mathcal{C},\ q > 0.$$

Proof. Exactness in augmented degrees $-1, 0$ is the sheaf condition for the basis. It implies the sheaf condition for all covers: refine a cover by a basis cover to glue, and use base changes of that basis cover to verify that the glued section restricts to the originally specified sections.

We first recall why an injective abelian sheaf $\mathcal{I}$ has exact augmented alternating Čech complex for a finite cover by monomorphisms. Let $\mathbb{Z}[U]$ denote the free abelian sheaf on the associated representable sheaf of U. The augmented chain complex

$$\cdots \longrightarrow \bigoplus_{i<j} \mathbb{Z}[U_i \times_U U_j] \longrightarrow \bigoplus_i \mathbb{Z}[U_i] \longrightarrow \mathbb{Z}[U] \longrightarrow 0$$

is exact. Locally over a section of U, one of the covering subobjects is available, and inserting its index contracts the augmented complex. This local exactness is exactness after sheafification. Applying the exact contravariant functor $\operatorname{Hom}(-, \mathcal{I})$, and using $\operatorname{Hom}(\mathbb{Z}[U], \mathcal{I}) = \mathcal{I}(U)$, proves the assertion. This is also the usual injective Čech-acyclicity statement; see [2, Tag 03AW].

Embed $\mathcal{E}$ in an injective abelian sheaf $\mathcal{I}$, and let $\mathcal{Q} = \mathcal{I}/\mathcal{E}$. Such embeddings exist on any site [2, Tag 01DP]. We claim that

$$0 \longrightarrow \mathcal{E}(U) \longrightarrow \mathcal{I}(U) \longrightarrow \mathcal{Q}(U) \longrightarrow 0 \tag{10}$$

is exact for every object U.

Indeed, a section of $\mathcal{Q}(U)$ lifts locally to $\mathcal{I}$, and we may choose a finite basis cover for these lifts. Local lifting for epimorphisms of abelian sheaves is part of the usual description of sheaf cokernels

[2, Tag 03CM]. The differences of the local lifts form a Čech 1-cocycle with values in $\mathcal{E}$. By the assumed exactness, this is a coboundary. Adjusting the lifts by a 0-cochain in $\mathcal{E}$ makes them agree, and the sheaf condition for $\mathcal{I}$ glues them to a global lift. This proves (10).

For every basis cover, (10) yields a short exact sequence of augmented Čech complexes. The complexes for $\mathcal{E}$ and $\mathcal{I}$ are exact, so the complex for $\mathcal{Q}$ is exact. Thus $\mathcal{Q}$ has the same property as $\mathcal{E}$.

The long exact sequence of the right derived functors of evaluation, applied to $0 \to \mathcal{E} \to \mathcal{I} \to \mathcal{Q} \to 0$, gives

$$H^1(U, \mathcal{E}) = 0, \qquad H^q(U, \mathcal{E}) \simeq H^{q-1}(U, \mathcal{Q}) \quad (q \geq 2).$$

Repeating the argument with $\mathcal{Q}$ proves vanishing in every positive degree. □

10 Proof of the theorem

Proof of Theorem 3.1. Let

$$\mathcal{G}_F = \mathcal{L}_P F.$$

Lemma 6.1 proves formula (1) and identifies its restriction to B_P with F.

Consider any object $P \to Q$. By Proposition 5.4, every birhizomic cover of this object has a refinement of the form

$$Q \longrightarrow Q\langle S_i \rangle, \qquad 1 \leq i \leq m,$$

which is a finite birhizomic cover. These finite rational covers form a basis, and their opposites are monomorphisms by Remark 4.4.

Define $H : B_Q \to \mathrm{Ab}$ by

$$H(Q_y) = \mathcal{G}_F(P \to Q \to Q_y).$$

Lemma 6.2 identifies $\mathcal{G}_F$, on monoids under Q, with $\mathcal{L}_Q H$. The displayed covering family is weakly valuative. Consequently, Proposition 8.2 proves exactness of its augmented alternating Čech complex with coefficients in $\mathcal{G}_F$.

This applies to every object Q and every basis cover. Lemma 9.1 therefore gives both

$$\mathcal{G}_F \in \mathrm{Sh}(A_P^{\mathrm{op}}, \mathrm{Ab}) \quad \text{and} \quad H^q(Q, \mathcal{G}_F) = 0 \quad (q > 0).$$

Since $\mathcal{G}_F$ is the presheaf left Kan extension and is already a sheaf, it is canonically $\pi^{-1}F$. This proves (i) and (ii).

Finally, the right adjoint π_* is restriction to B_P. Indeed, restriction is right adjoint to presheaf left Kan extension, and sheafification is left adjoint to inclusion of sheaves into presheaves. The topology on B_P^{op} is trivial, so no additional sheafification is involved on that side.

Take an injective resolution $\mathcal{G}_F \to \mathcal{I}^\bullet$ in abelian sheaves on A_P^{op}. Because cohomology in the presheaf category on B_P is computed objectwise, for every $x \in P$ we have

$$\big(\mathrm{R}^q \pi_* \mathcal{G}_F\big)(P_x) = H^q\big(\mathcal{I}^\bullet(P_x)\big) = H^q(P_x, \mathcal{G}_F).$$

These groups vanish for $q > 0$. In degree zero, the canonical left Kan extension unit gives

$$(\pi_* \mathcal{G}_F)(P_x) = \mathcal{G}_F(P_x) = F(P_x).$$

Thus the derived unit induces the identity isomorphism in degree zero and has zero higher cohomology on its target. It is therefore a quasi-isomorphism:

$$\boxed{F \xrightarrow{\sim} \mathrm{R}\pi_* \pi^{-1} F}.$$

□

References

[1] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Theorems 2.10 and 2.27.

[2] The Stacks Project Authors, *The Stacks Project*. Tags `00DB` (exactness of filtered colimits), `012X` (spectral sequences of double complexes), `01D6` (injective abelian groups), `01DP` (enough injectives for abelian sheaves on a site), `03CM` (abelian sheaves and local exactness), and `03AW` (injective Čech acyclicity).